LC29H&LC79H Series

Low Power Mode Application Note

GNSS Products

Version: 1.0

Date: 2026-02-03

Status: Released

At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:

Quectel Wireless Solutions Co., Ltd.
No. 8 Waipojing Road, Sijing Town, Songjiang District, Shanghai 201601, China
Tel: +86 21 5108 6236
Email: info@quectel.com

Or our local offices. For more information, please visit:
https://www.quectel.com/contact/.

For technical support, or to report documentation errors, please visit:
https://www.quectel.com/tech-support/.
Or email us at: support@quectel.com.

Legal Notices

We provide this document to support your product design. You are required to design your products based on the specifications and parameters set forth herein. You agree that you are responsible for using independent analysis and evaluation in designing intended products, and we provide reference designs for illustrative purposes only. Before using any hardware, software or service guided by this document, please read this notice carefully. Even though we employ commercially reasonable efforts to provide the best possible experience, you hereby acknowledge and agree that this document and related services hereunder are provided to you on an “as available” basis. You acknowledge and agree that we may add to, amend, or restate this document at any time at our sole discretion without any prior notice to you, and such additions, amendments, or restatements shall be binding upon you.

Use and Disclosure Restrictions

License Agreements

The recipient of any hardware, software, materials, or documentation provided by us shall keep such content confidential, unless expressly authorized by us. The recipient shall not disclose, access, or use any part of the received content for any purpose other than the execution and implementation of the intended project.

Copyright

Our and third-party products hereunder may contain copyrighted materials, including but not limited to protected content, hardware, software, and documentation owned by us or applicable third parties. Unless prior written consent is obtained, you shall not access, use, or disclose any documents or information provided by us, nor shall you copy, reproduce, republish, display, translate, distribute, merge, modify, or create derivative works from any such copyrighted materials. We and the applicable third party retain exclusive rights to all copyrighted materials. No license to any patents, copyrights, trademarks, or service marks shall be granted or transferred. For the avoidance of doubt, no form of purchase shall be construed as granting any license beyond a normal, non-exclusive, royalty-free license to use the product. We reserve the right to pursue legal action against any violation of confidentiality

obligations, unauthorized use, or any other unlawful or malicious use of the aforementioned documents and information.

Trademarks

Unless otherwise expressly provided, nothing in this document shall be construed as conferring any rights to use any trademark, trade name, name, abbreviation, or counterfeit thereof owned by us or any third party in advertising, publicity, or any other contexts.

Third-Party Rights

You understand that this document may refer to hardware, software, and/or documentation owned by one or more third parties ("third-party materials"). Use of such third-party materials is subject to all applicable restrictions and obligations set forth herein.

We make no warranty or representation, either express or implied, regarding the third-party materials, including but not limited to any implied or statutory, warranties of merchantability or fitness for a particular purpose, quiet enjoyment, system integration, information accuracy, and non-infringement of any third-party intellectual property rights with regard to the licensed technology or use thereof. Nothing herein constitutes a representation or warranty by us to either develop, enhance, modify, distribute, market, sell, offer for sale, or otherwise maintain production of any our products or any other hardware, software, device, tool, information, or product. We moreover disclaim any and all warranties arising from the course of dealing, course of performance, or usage of trade.

Privacy Policy

To enable product functionality, certain device data may be uploaded to our or third-party servers, including those operated by carriers, chipset suppliers, or servers designated by you. We strictly comply with applicable laws and regulations and will retain, use, disclose, or otherwise process relevant data solely for the purpose of enabling product functionality, or as permitted by applicable laws. Before interacting with any third party regarding data exchange, please be informed of and understand their privacy and data security policies.

Disclaimer

a) We shall not be liable for any damages resulting from failure to comply with applicable operational or design specifications.
b) We shall bear no liability for any inaccuracies or omissions in this document, nor for any damages arising from the use of the information contained herein.
c) While we make every effort to ensure the integrity, accuracy, and timeliness of the features and functions under development, errors or omissions may nevertheless occur. Unless otherwise provided in a valid written agreement, we make no warranties of any kind, express, implied, or statutory, and disclaim all liability for any loss or damage arising from the use of any features or functions under development, to the maximum extent permitted by law, regardless of whether such loss or damage is foreseeable.
d) We assume no legal responsibility for the accessibility, safety, accuracy, availability, legality, or completeness of any information, content, advertising, commercial offers, products, services, or materials on third-party websites or third-party resources.

Copyright © Quectel Wireless Solutions Co., Ltd. 2026. All rights reserved.

About the Document

Document Information	
Title	**LC29H&LC79H Series Low Power Mode Application Note**
Subtitle	GNSS Products
Document Type	Application Note
Document Status	Released

Revision History

Version	Date	Description
-	2025-10-29	Creation of the document
1.0	2026-02-03	First official release

Contents

About the Document **3**
Contents **4**
Table Index **5**
Figure Index **6**

1 Introduction **7**

2 Low Power Modes **11**
2.1. Backup Mode 11
2.1.1. Operation Mechanism 11
2.1.2. Testing Procedure 12
2.2. Periodic Power Saving Mode 13
2.2.1. Operation Mechanism 13
2.2.2. Testing Procedure 15
2.3. ALP Mode 16
2.3.1. Operation Mechanism 16
2.3.1.1. Comparison Between ALP Mode and Continuous Mode 18
2.3.1.2. Shorten Time to Achieve Low Power Consumption in ALP Mode 18
2.3.2. Testing Procedure 19
2.4. FLP Mode 20
2.4.1. Operation Mechanism 20
2.4.2. Testing Procedure 21

3 EPH and ALM Related Messages **22**
3.1. EPH Related Messages 22
3.1.1. PAIR530: PAIR_EPH_GET_STATUS 22
3.1.2. PAIR531: PAIR_EPH_SET_DATA 24
3.1.3. PAIR532: PAIR_EPH_GET_DATA 36
3.1.4. PAIR533: PAIR_EPH_CLEAR 40
3.1.5. PAIR534: PAIR_EPH_NOTIFY_ENABLE 41
3.1.6. PAIR535: PAIR_EPH_NOTIFY 41
3.2. ALM Related Messages 42
3.2.1. PAIR550: PAIR_ALM_GET_STATUS 42
3.2.2. PAIR551: PAIR_ALM_SET_DATA 44
3.2.3. PAIR552: PAIR_ALM_GET_DATA 50
3.2.4. PAIR553: PAIR_ALM_CLEAR 53

4 Appendix A References **55**
5 Appendix B Special Characters **57**

Table Index

Table 1: Commands Related to Low Power Modes 7
Table 2: Navigation Modes 8
Table 3: Low Power Modes and Corresponding Configurations 9
Table 4: System Resources Utilized in Low Power Modes 11
Table 5: Test Environment in Backup Mode 12
Table 6: Test Environment in Periodic Power Saving Mode 15
Table 7: Module Performance in Continuous Mode and ALP Mode 18
Table 8: Test Environment in ALP Mode – Entering ALP Mode After Power On 19
Table 9: Related Documents 55
Table 10: Terms and Abbreviations 55
Table 11: Special Characters 57

Figure Index

Figure 1: Measured Power Consumption in Backup Mode 13
Figure 2: Module Operation in Smart Periodic Mode 14
Figure 3: Module Operation in Strict Periodic Mode 15
Figure 4: Measured Power Consumption in Periodic Power Saving Mode 16
Figure 5: Module Operation in ALP Mode 17
Figure 6: Power Consumption Measured in ALP Mode After Power On 20
Figure 7: Module Operation in FLP Mode 20
Figure 8: EPH Data Format – GPS (L1) and QZSS (L1) 28
Figure 9: EPH Data Format – GPS (L5) and QZSS (L5) 29
Figure 10: EPH Data Format – GLONASS (L1) 30
Figure 11: EPH Data Format – Galileo (E1) 31
Figure 12: EPH Data Format – Galileo (Ea5) 32
Figure 13: EPH Data Format – BDS GEO (B1I) 33
Figure 14: EPH Data Format – BDS MEO (B1I) 34
Figure 15: EPH Data Format – BDS MEO (B2a) 35
Figure 16: ALM Data Format – GPS (L1) and QZSS (L1) 47
Figure 17: ALM Data Format – GPS (L5) and QZSS (L5) 48
Figure 18: ALM Data Format – GLONASS (L1) 48
Figure 19: ALM Data Format – Galileo (E1) 49
Figure 20: ALM Data Format – Galileo (E5a) 49
Figure 21: ALM Data Format – BDS (B1I) 49
Figure 22: ALM Data Format – BDS (B2a) 50

1 Introduction

The LC29H series and LC79H series modules support different low power modes with enhanced power consumption and outstanding positioning performances. This document describes the operation mechanisms and related testing procedures of Backup, Periodic Power Saving, Adaptive Low Power (ALP) and Fitness Low Power (FLP) modes, as well as EPH and ALM related messages.

In Backup mode, the module is in an inactive state where most functions are disabled, and the power consumption mainly comes from the RTC circuit.

Periodic Power Saving mode consists of running (active) and sleeping stages, whose time span can be adjusted to reduce power consumption.

ALP mode is a subset of FLP mode that differs in navigation mode. The differences between FLP and ALP modes are detailed in *Table 3: Low Power Modes and Corresponding Configurations*.

The commands related to low power modes are listed below. Before implementing any of the commands below, refer to *document [1] protocol specification*.

Table 1: Commands Related to Low Power Modes

Message	Description
PQTMCFGFIXRATE PAIR050/PAIR051	Sets/Gets the position fix interval.
PQTMCFGCNST PAIR066/PAIR067	Sets/Gets the GNSS constellation configuration.
PQTMCFGNAVMODE PAIR080/PAIR081	Sets/Gets the navigation mode.
PQTMBKP PAIR650	Shuts down the GNSS system, except the clock. This command sets the CPU core to Backup mode, in which it cannot receive any commands.
PQTMCFGLPM	Set/Gets low power mode. Configurable low-power modes include FLP, ALP and periodic power saving modes.

Table 2: Navigation Modes

Navigation Mode	Description
Normal mode	Used for general purposes.
Fitness mode	Used for running and walking purposes, making low-speed movement (< 5 m/s) more impactful on position calculation.
Swimming mode	Used for swimming purposes to smooth the trajectory and improve the accuracy of distance calculation.
Dynamic Flight mode	Used for drone applications with equivalent dynamic range and vertical acceleration at different flight phases (for example, hovering and cruising).
Balloon mode	Used for high-altitude balloon scenarios where the vertical movement has a greater impact on the position calculation.
Stationary mode	Used for stationary applications where a zero dynamic is assumed.

The corresponding configurations of the low power modes supported by LC29H and LC79H series are as follows:

Table 3: Low Power Modes and Corresponding Configurations

Low Power Mode	Navigation Mode	Constellation Option	Fix Update Rate	Supported Module
• Backup mode • Periodic Power Saving mode	• Normal mode • Fitness mode • Swimming mode • Dynamic Flight mode • Balloon mode • Stationary mode	• GPS + GLONASS + Galileo + BDS + NavIC + QZSS • GPS + GLONASS + Galileo + BDS + QZSS • GPS + GLONASS + Galileo + QZSS • GPS + GLONASS + BDS + QZSS • GPS + Galileo + NavIC + QZSS • GPS + Galileo + BDS + QZSS • GPS + GLONASS + QZSS • GPS + Galileo + QZSS • GPS + BDS + QZSS • GPS + QZSS • BDS • NavIC	• 1 Hz • 2 Hz • 5 Hz • 10 Hz	LC29H (AA) and LC79H (AL)
Backup mode	Normal mode	• GPS + GLONASS + Galileo + BDS + NavIC + QZSS • GPS + GLONASS + Galileo + BDS + QZSS • GPS + Galileo + NavIC + QZSS • GPS + Galileo + BDS + QZSS • GPS + Galileo + QZSS • GPS + BDS + QZSS • GPS + QZSS • BDS • NavIC	• 1 Hz • 2 Hz • 5 Hz • 10 Hz	LC29H (BA, CA)

Low Power Mode	Navigation Mode	Constellation Option	Fix Update Rate	Supported Module
		● GPS + GLONASS + Galileo + BDS + QZSS ● GPS + Galileo + BDS + QZSS ● GPS + Galileo + QZSS ● GPS + BDS + QZSS ● GPS + QZSS ● BDS	● 1 Hz ● 2 Hz ● 5 Hz ● 10Hz	LC29H (EA) and LC79H (EL)
			1 Hz	LC29H (DA)
ALP mode	● Normal mode ● Fitness mode	● GPS + GLONASS + Galileo + BDS + NavIC + QZSS ● GPS + GLONASS + Galileo + BDS + QZSS ● GPS + GLONASS + Galileo + QZSS ● GPS + GLONASS + BDS + QZSS ● GPS + Galileo + NavIC + QZSS ● GPS + Galileo + BDS + QZSS ● GPS + GLONASS + QZSS ● GPS + Galileo + QZSS ● GPS + BDS + QZSS ● GPS + QZSS ● BDS ● NavIC	1 Hz	LC29H (AA) and LC79H (AL)
FLP mode	Fitness mode	GPS + GLONASS + Galileo + BDS	1 Hz	LC29H (AA) and LC79H (AL)

NOTE

1. The navigation mode, GNSS satellite configuration, and fix update rate are the prerequisites for the operation of the low power modes. If the module is not in the tracking mode that matches one of the available navigation profiles, it cannot support low power modes.
2. In the above table, QZSS constellation can be enabled or disabled in the constellation configurations supporting QZSS.

2 Low Power Modes

To extend battery operation time and meet customer requirements, the module supports several low power modes.

Table 4: System Resources Utilized in Low Power Modes

System Resource	Backup	Periodic Power Saving (Running Stage)	Periodic Power Saving (Sleeping Stage)	ALP	FLP
RTC Macro	●	●	●	●	●
RTC SRAM	●	●	●	●	●
Internal Circuits	-	●	●	●	●
CPU core	-	●	●	●	●
SRAM	-	●	●	●	●
TCXO	-	●	●	●	●
GPIO	-	●	●	●	●
RF	-	●	-	●	●

2.1. Backup Mode

2.1.1. Operation Mechanism

Backup mode is used to improve power efficiency. It shuts down most system components, such as RF controller, processing engine DSP, CPU core and so on. As a result, while the module is in Backup mode, the module's communication port is unresponsive to any incoming commands and no data is transmitted over it. Before it enters the Backup mode, the module saves navigation data, including ephemeris, almanac, location and time to RTC SRAM for the next navigation cycle to shorten the TTFF. The CPU core can be woken up by setting a timer or by first powering up the VCC pin and then pulling the WAKEUP pin high. For details, refer to *document [1] protocol specification* and *documents [2] and [3] hardware designs*.

Once the CPU core exits the Backup mode, the navigation data will be loaded from RTC SRAM. If all the data is valid, a shorter TTFF can be achieved.

2.1.2. Testing Procedure

To Enter the Backup mode (constellation configuration: GPS + GLONASS + Galileo + BDS + QZSS):

1. Send **PQTMBKP** command;
2. Disconnect VCC power supply and keep the V_BCKP pin powered.

To exit the Backup mode:

- If **<Second>** is set to 0, restore the VCC power supply first and then pull the WAKEUP pin high for at least 10 ms.
- If **<Second>** is not 0 and the timer has not expired, restore the VCC power supply first and then pull the WAKEUP pin high for at least 10 ms.
- If **<Second>** is not 0 and the timer expires, restore the VCC power.

Table 5: Test Environment in Backup Mode

Item	Description
Module	LC29H (AA)
Voltage	3.3 V
RF Signal Strength	-130 dBm
Equipment	Keysight 66319B
Command	**$PQTMBKP,0*5D**
Execution Time	After the module gets a fix.
Axis	Vertical axis: current drain (μA) Horizontal axis: run time

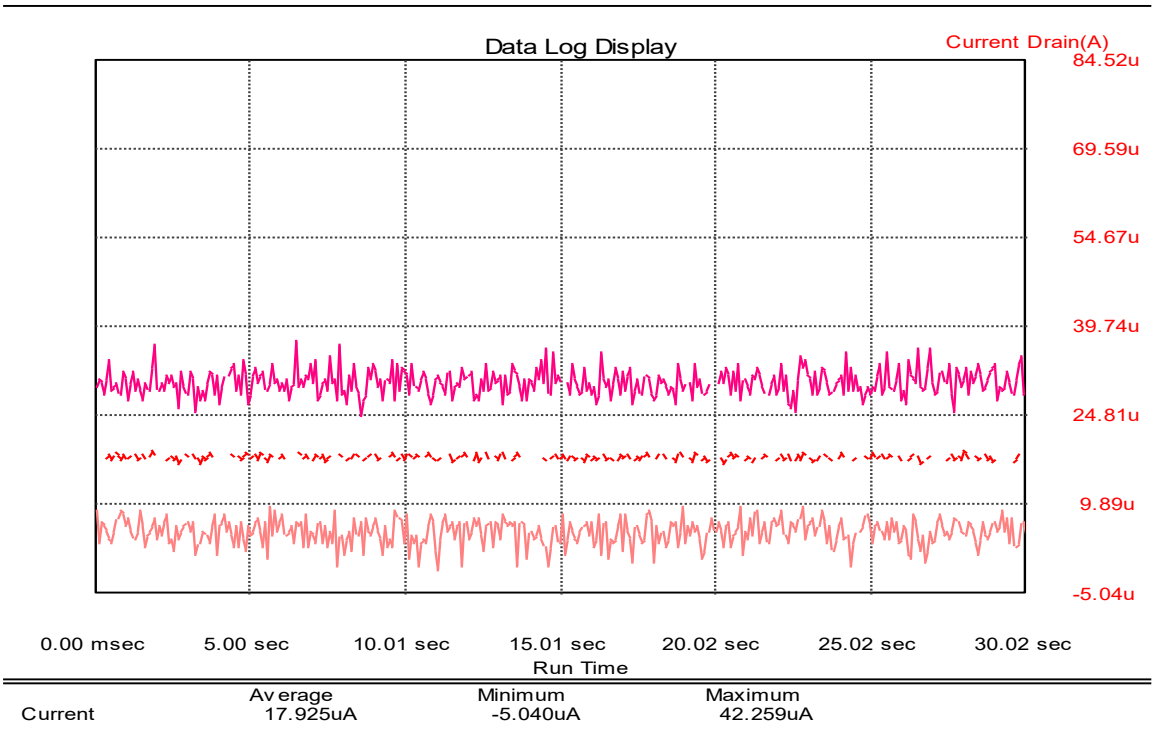


Figure 1: Measured Power Consumption in Backup Mode

2.2. Periodic Power Saving Mode

2.2.1. Operation Mechanism

In the Periodic Power Saving mode, power consumption can be reduced by turning off the DSP and periodically putting the CPU core to sleep. Average consumption is conditioned by the duration of running and sleeping times, which can be adjusted as needed.

- During the running time, the DSP and CPU core are active and perform their respective tasks, while the GNSS module generates the position-related information.
- During the sleeping time, the DSP is paused while the CPU core is in the sleeping stage.

The time span of the running (active) and sleeping stages can be dynamically changed or strictly limited, i.e., Smart Periodic Mode and Strict Periodic Mode. In Smart Periodic mode, during the running stage, the module observes the current satellite signal status, satellite health status, collected ephemeris status, etc., to determine the time needed to collect ephemeris data. Therefore, when **PAIR690** is implemented, the system will automatically extend the running time to collect the required ephemeris data, increasing the likelihood that the position will be available immediately after the module wakes up. To shorten TTFF after waking up, it is recommended to use Smart Periodic mode to dynamically change the time span of the running stage to obtain more navigation information, such as pseudorange, ephemeris, and almanac. In Strict Periodic mode the duration of every cycle is fixed because the time span of the running stage is

limited to a specific value. This fixed time span means that TTFF could potentially be affected by the fact that the module may not be able to complete a satellite data update, such as ephemeris, within a single cycle.

Periodic pattern schematic diagram with different colors representing specific stages:

- Blue: module is in running (active) stage and the GNSS signal is good.
- Red: module is in sleeping stage and the GNSS signal is good.
- Yellow: module is in running (active) stage and there is no GNSS signal.
- Green: module is in sleeping stage and there is no GNSS signal.
- Brown: the extension time of running (active) stage.

Module operation in Smart Periodic mode is illustrated in the following figure:

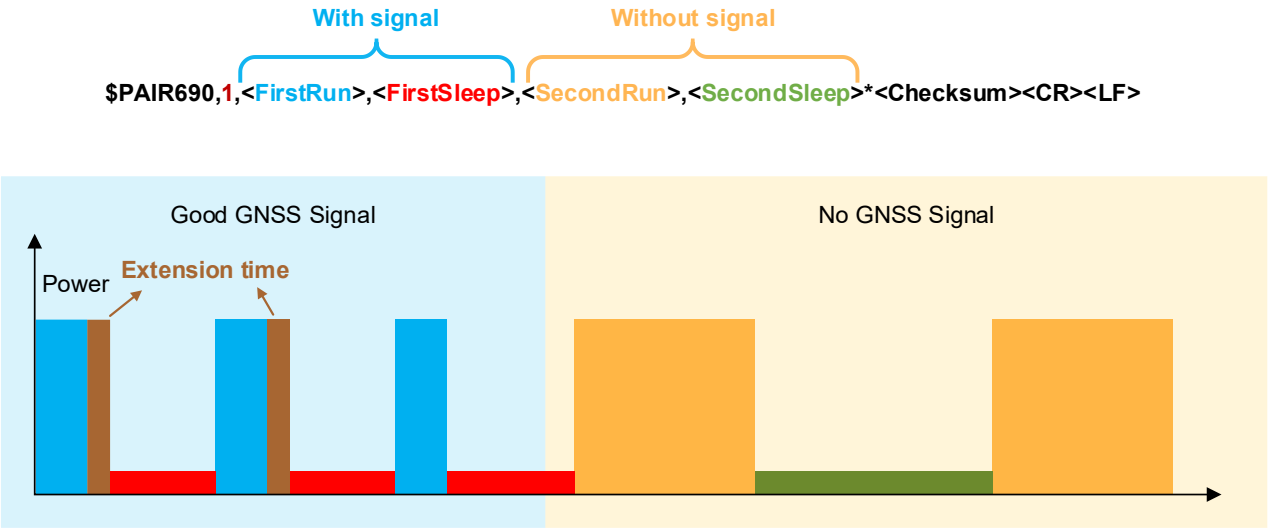


Figure 2: Module Operation in Smart Periodic Mode

Module operation in Strict Periodic mode is illustrated in the following figure:

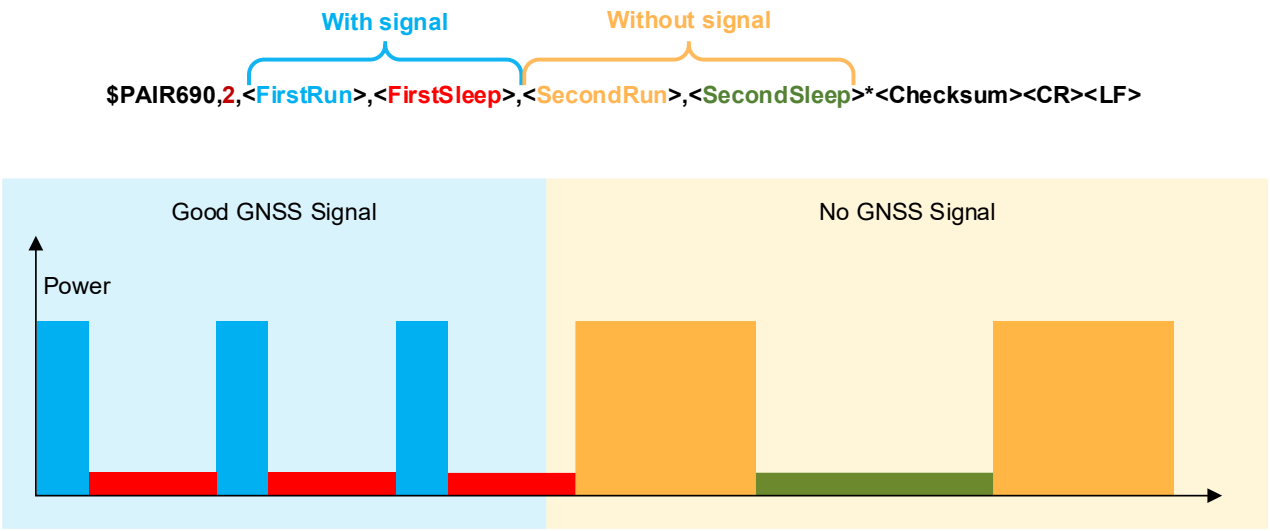


Figure 3: Module Operation in Strict Periodic Mode

2.2.2. Testing Procedure

To enter the Periodic Power Saving mode (constellation configuration: GPS + GLONASS + Galileo + BDS + QZSS):

1. Power on the module.
2. Send **PQTMCFGLPM**.

The module returns **$PQTMCFGLPM,OK*23** when it enters the Periodic Power Saving mode.

To exit the Periodic Power Saving mode:
Send **$PQTMCFGLPM,W,0*6C**.

Table 6: Test Environment in Periodic Power Saving Mode

Item	Description
Module	LC29H (AA)
Voltage	3.3 V
RF Signal Strength	-130 dBm
Equipment	Keysight 66319B
Command	**$PQTMCFGLPM,W,4,2,10,30,30,40*73**
Execution Time	After the module gets a fix.

Item	Description
Axis	Vertical axis: current drain (mA) Horizontal axis: run time

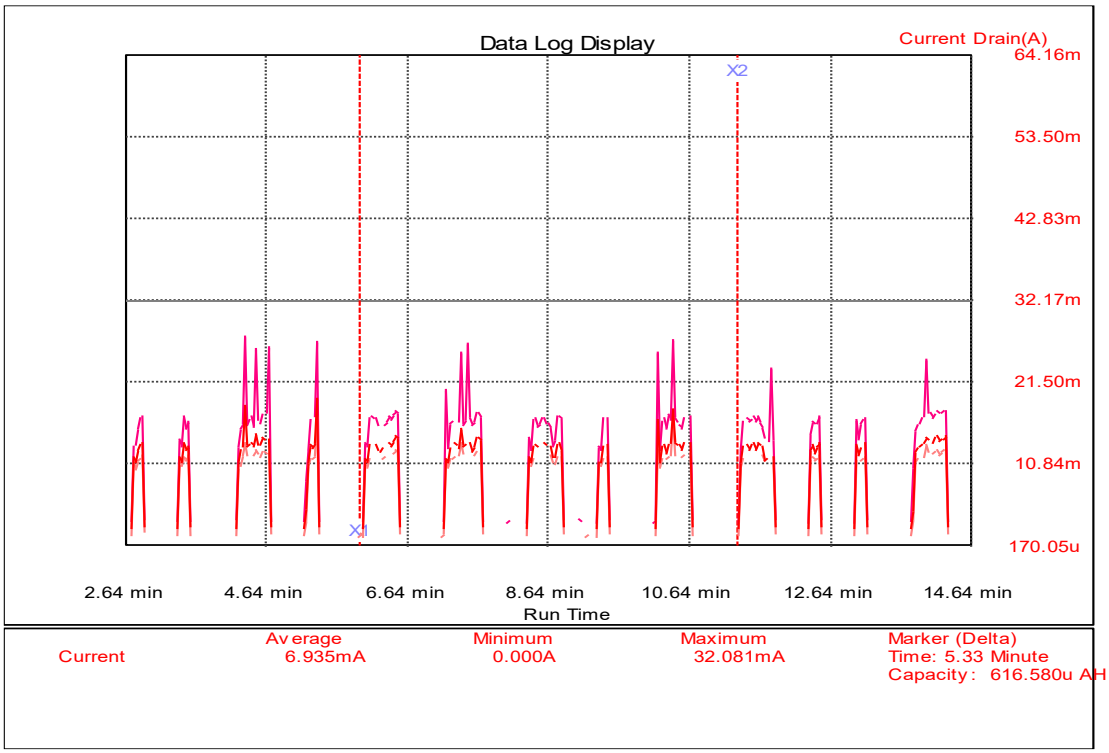


Figure 4: Measured Power Consumption in Periodic Power Saving Mode

2.3. ALP Mode

2.3.1. Operation Mechanism

In the ALP mode, power is saved by using a method similar to duty cycle, but the position quality is degraded compared to that in the tracking mode (in the Continuous mode). The receiver outputs a positioning solution at every epoch, but with lower precision. The ALP mode is suitable for scenarios where long-term module operation and power efficiency are critical. However, it is not suitable for scenarios where high positioning accuracy in short-term operation are prioritized.

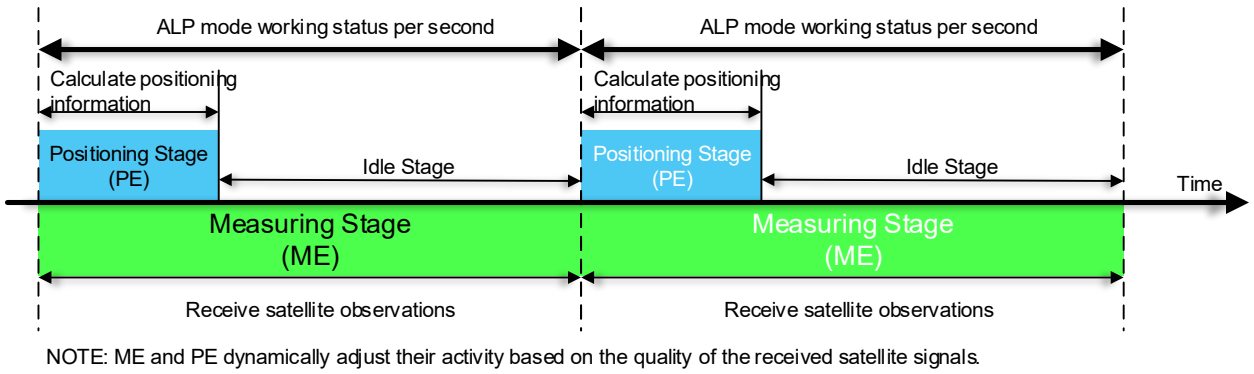


Figure 5: Module Operation in ALP Mode

As illustrated above, the ALP mode consists of three stages:

- Measuring Stage: ME receives satellite observations.
- Positioning Stage: PE calculates positioning information.
- Idle Stage: Module is in idle stage.

After initial power-on and entering the ALP mode, the module does not immediately reduce power consumption. Instead, it uses the ME and PE to actively search for all satellites in view, ensuring the shortest possible TTFF and the highest sensitivity for satellite signal reception. As the module downloads enough ephemeris and almanac data, it gradually reduces power consumption. Once both the ephemeris and almanac data are fully downloaded, the power consumption of the module stabilizes at its optimal level for the ALP mode. The detailed process is as follows:

1. To achieve a good positioning performance after startup, the module continuously searches for GNSS satellite signals, and its ME and PE consume most of the power.

2. As the module accumulates enough GNSS satellite data, it gradually reduces the usage of ME and PE, resulting in a gradual reduction in power consumption.

3. After downloading the complete almanac (which takes about 12.5 minutes) the power consumption of the module stabilizes. The ME and PE then dynamically adjust their activity based on the quality of the received satellite signals to maintain position accuracy while optimizing power usage.

In the ALP mode, the module searches for satellites and adjusts the search time according to the actual conditions to achieve better positioning results while minimizing power consumption. This means that the satellite search time can be extended when satellite availability and signal quality are weak and the satellite search time can be shortened when satellite availability and signal quality are good.

2.3.1.1. Comparison Between ALP Mode and Continuous Mode

The following table compares the ALP mode with the Continuous mode.

Table 7: Module Performance in Continuous Mode and ALP Mode

Feature	Continuous Mode	ALP Mode
Power Consumption	High	Medium
Positioning Accuracy	High	Medium
Satellite Search	No limitations	Slightly limited
RF Circuit Activity	Unrestricted	Unrestricted

2.3.1.2. Shorten Time to Achieve Low Power Consumption in ALP Mode

The module reduces power consumption in ALP mode when the following conditions are met:

- A valid position is obtained.
- The entire almanac has been downloaded (about 12.5 minutes).
- The ephemeris for each satellite in view is valid.

To shorten the time to achieve low power consumption in ALP mode, consider these options:

- **Save ephemeris and almanac data to flash.** Send **$PAIR511*3F** to save the ephemeris and almanac data to flash memory after each satellite search or before the module is turned off, to avoid data loss due to expired ephemeris data or a power failure.

- **Use the Backup mode instead of powering down the module.** The module saves satellite data to RTC RAM in the Backup mode. By waking the module from the Backup mode and then entering the ALP mode (provided the satellite data is still valid), the time required to achieve low power consumption in the ALP mode can be reduced.

- **Use external ephemeris and almanac data.** If another device has valid ephemeris and almanac data for the satellites in view, send the data to the module to significantly reduce the time for achieving low power consumption in ALP mode. For relevant commands, refer to *Chapter 3 EPH and ALM Related Messages.*

- **Inject AGNSS EPO files.** Injecting EPO files into the module using the AGNSS service can also reduce the time for achieving low power consumption in ALP mode. This is an auxiliary method that may not be as efficient as the other options. For details on EPO file injection, refer to *document [4]*

AGNSS application note.

2.3.2. Testing Procedure

Please refer to *Table 3: Low Power Modes and Corresponding Configurations* before performing the following test.

To enter the ALP mode (constellation configuration: GPS + GLONASS + Galileo + BDS + QZSS):
1. Power on the module.
2. Send **PQTMCFGNAVMODE** to set the navigation mode to Normal mode.
3. Send **$PQTMCFGLPM,W,1,2*73** to enter the ALP mode.

The module returns **$PQTMCFGLPM,OK*23** when it enters the ALP mode.

To exit the ALP mode:
Send **$PQTMCFGLPM,W,0*6C**.

Table 8: Test Environment in ALP Mode – Entering ALP Mode After Power On

Item	Description
Module	LC29H (AA)
Voltage	3.3 V
RF Signal Strength	-130 dBm
Equipment	Keysight 66319B
Command	**$PQTMCFGLPM,W,1,2*73**
Execution Time	Immediately after the module is powered on.
Axis	Vertical axis: current drain (mA) Horizontal axis: run time

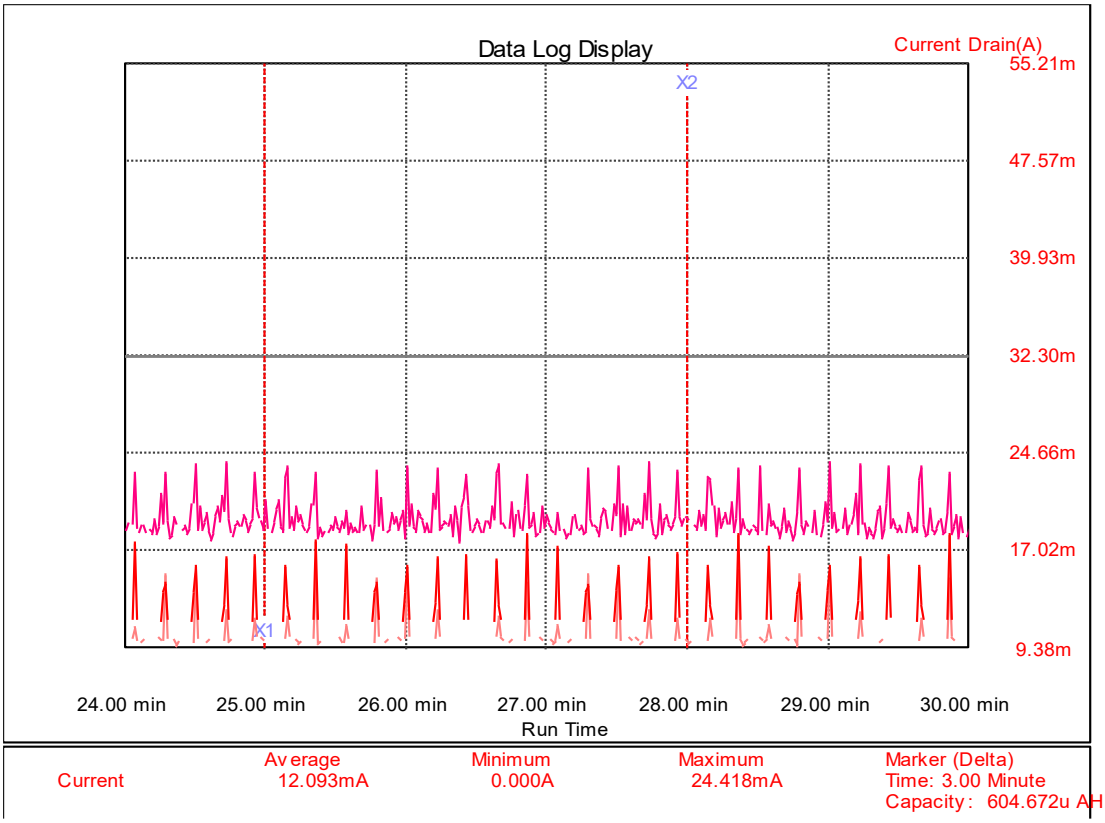


Figure 6: Power Consumption Measured in ALP Mode After Power On

2.4. FLP Mode

2.4.1. Operation Mechanism

In FLP mode, power is saved by using a method similar to duty cycle, but the position quality is degraded compared to that in tracking mode (in the Continuous mode).

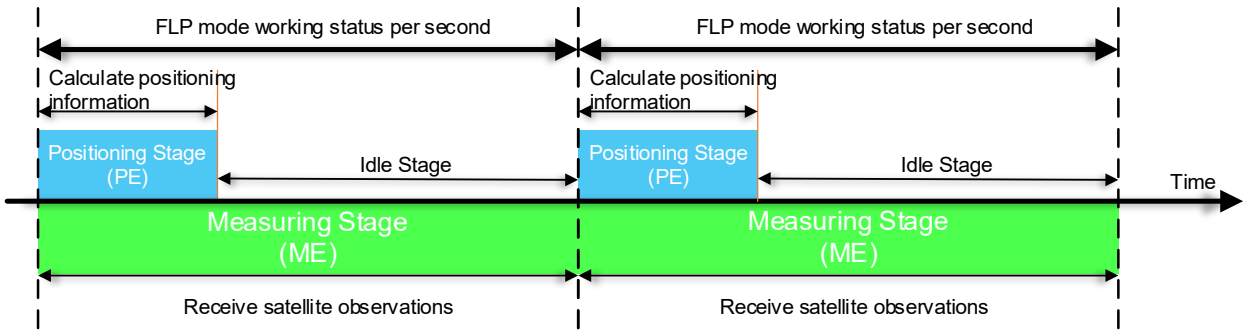


Figure 7: Module Operation in FLP Mode

As shown above, the FLP mode consists of three stages:

- Measuring Stage: ME receives satellite observations.
- Positioning Stage: PE calculates positioning information.
- Idle Stage: the module is in idle stage.

Due to the satellite search limitations in FLP mode, the positioning accuracy is lower compared to that in the Continuous mode. However, because the module limits the number of satellites it searches, the time for satellite search and calculation is reduced, resulting in lower power consumption in this mode.

2.4.2. Testing Procedure

Please refer to *Table 3: Low Power Modes and Corresponding Configurations* before performing the following test.

To enter the FLP mode (constellation configuration: GPS + GLONASS + Galileo + BDS):

1. Power on the module.
2. Send **PQTMCFGNAVMODE** to set the navigation mode to Fitness mode.
3. Send **$PQTMCFGLPM,W,2,2*70** to enter the FLP mode.

The module returns **$PQTMCFGLPM,OK*23** when it enters the FLP mode.

To exit the FLP mode:
Send **$PQTMCFGLPM,W,0*6C**.

The module's power consumption in FLP mode is similar to that in ALP mode. For the description of power consumption in the ALP mode, please refer to *Chapter 2.3.2 Testing Procedure*.

3 EPH and ALM Related Messages

This chapter explains the EPH and ALM related messages supported by the module.

NOTE

For details about **PAIR001** message, see *document [1] protocol specification*.

3.1. EPH Related Messages

3.1.1. PAIR530: PAIR_EPH_GET_STATUS

Queries the satellites whose valid EPH data will be available for a specified future time period.

Type:

Command

Synopsis:

```
$PAIR530,<Constellation>,<Time_Interval>*<Checksum><CR><LF>
```

Parameter:

Field	Format	Unit	Description
<Constellation>	Numeric	-	GNSS System ID. 0 = GPS 1 = GLONASS 2 = Galileo 3 = BDS 4 = QZSS
<Time_Interval>	Numeric	Second	Specified future time period. Range: 1–7200 (2 hours).

Result:

Returns **PAIR001** message and the query result.

Query result message format:

```
$PAIR530,<Constellation>,<L1_SV>[,<L5_SV>]*<Checksum><CR><LF>
```

Parameters included in the result:

Field	Format	Unit	Description
<Constellation>	Numeric	-	GNSS System ID. 0 = GPS 1 = GLONASS 2 = Galileo 3 = BDS 4 = QZSS
<L1_SV>	Hexadecimal	-	Valid EPH SV ID (L1 band). Field Lengths for different constellations (in ASCII format) are as follows. GPS: 8 bytes GLONASS: 6 bytes Galileo: 16 bytes BDS: 16 bytes QZSS: 8 bytes Note that each bit represents a satellite, and SV ID counting starts from the least significant bit. Use GPS as an example. If Bit 0 is 1, the EPH data of GPS satellite 1 is available.
<L5_SV>	Hexadecimal	-	Valid EPH SV ID (L5 band). When **<Constellation>** = 1, this field is omitted. Field Lengths for different constellations (in ASCII format) are as follows. GPS: 8 bytes GLONASS: 6 bytes Galileo: 16 bytes BDS: 16 bytes QZSS: 8 bytes Note that each bit represents a satellite, and SV ID counting starts from the least significant bit. Use GPS as an example. If Bit 0 is 1, the EPH data of GPS satellite 1 is available.

Example:

```
$PAIR530,0,1800*05
$PAIR001,530,1*3C
$PAIR001,530,0*3D
```

$PAIR530,0,24BD3DEF,209025AD*5D
//The binary format of 24BD3DEF is 0010 0100 1011 1101 0011 1101 1110 1111. The corresponding satellite numbers are 1, 2, 3, 4, 6, 7, 8, 9, 11, 12, 13, 14, 17, 19, 20, 21, 22, 24, 27, 30.
// The binary format of 209025AD is 0010 0000 1001 0000 0010 0101 1010 1101. The corresponding satellite numbers are 1, 3, 4, 6, 8, 9, 11, 14, 21, 24, 30.

3.1.2. PAIR531: PAIR_EPH_SET_DATA

Sends EPH data to GNSS module.

Type:

Input

Synopsis:

$PAIR531,<Constellation>,<Signal_ID>,<SV_ID>,<EPH_Data>*<Checksum><CR><LF>

Parameter:

Field	Format	Unit	Description
<Constellation>	Numeric	-	GNSS System ID. 0 = GPS 1 = GLONASS 2 = Galileo 3 = BDS 4 = QZSS
<Signal_ID>	Numeric	-	Signal ID. 0 = L1 (including Galileo E1 and BDS B1I) 1 = L5 (including GPS/QZSS L5, Galileo E5a and BDS B2a)
<SV_ID>	Numeric	-	SV ID. Ranges for different constellations: GPS: 1–32 GLONASS: 1–24 Galileo: 1–36 BDS: 1–63 (GEO: 1–5, 59–63; MEO: 6–58) QZSS: 1–7
<EPH_Data>	Hexadecimal	-	EPH data. EPH data format for different constellations: **GPS (L1) and QZSS (L1):** Word[0]–Word[23]. 24 words of EPH data from words 3–10 of subframes 1–3 in the GPS or QZSS Navigation Message. The 30-bit raw data of each word have been logically right-shifted by 6 bits to remove the 6 parity bits, leaving 24 bits of valid data per word. For details on data format, see *Figure 8: EPH Data Format – GPS (L1) and QZSS (L1)*.

Field	Format	Unit	Description
			GPS (L5) and QZSS (L5): Word[0]–Word[19]. 20 words of the ephemeris data consisting of bits 33–276 from message types 10, 11 and 30 in the Navigation Message. Every word contains 32 bits, and eight words make up one message type. The last four words contain the clock data. For details on data format, see *Figure 9: EPH Data Format – GPS (L5) and QZSS (L5)*. **GLONASS (L1):** Word[0]–Word[15]. Word[0]–Word[14] of EPH data are from strings 1–5 of a frame in a GLONASS Navigation Message. Word[15] indicates the frequency channel (range: 1–14). For details on data format, see *Figure 10: EPH Data Format – GLONASS (L1)*. **Galileo (E1):** Word[0]–Word[18]. 19 words of EPH data in the Galileo Navigation Message include 16 words from subframes 1–4, and 3 words from subframe 5, which include BGD, signal health status, data validity, and GST. Note that only Word[0]–Word[15] need to be input when sending ephemeris, while Word[0]–Word[18] will be provided when receiving ephemeris. The Page Type (6-bit) and IOD_{nav} (10-bit) fields have been removed and the remaining bits of each word are right-shifted. For details on data format, see *Figure 11: EPH Data Format – Galileo (E1)*. **Galileo (E5a):** Word[0]–Word[31]. 32 words of ephemeris data from word types 1–4 in subframes of the Galileo E5a Navigation Message. Every word contains 32 bits, and eight words make up one word type. The first word of a word type contains bits 217–248. The second word of a word type contains bits 185–216, and so on. …. The eighth word of a word type contains bits 1–24. For details on data format, see *Figure 12: EPH Data Format – Galileo (Ea5)*. **BDS GEO (B1I):** Word[0]–Word[29]. 30 words of EPH data from pages 1–10 of subframe 1 in the BDS GEO D2

Field	Format	Unit	Description
			Navigation Message. The 30-bit raw data of each word have been logically right-shifted by 8 bits to remove the 8 parity bits. For details on data format, see *Figure 13: EPH Data Format – BDS GEO (B1I)*. **BDS MEO (B1I):** Word[0]–Word[20]. 21 words of EPH data from subframes 1–3 in the BDS MEO D1 Navigation Message. The 30-bit raw data of each word have been logically right-shifted by 8 bits to remove the 8 parity bits. For details on data format, see *Figure 14: EPH Data Format – BDS MEO (B1I)*. **BDS MEO (B2a):** Word[0]–Word[20]. 21 words of ephemeris subframe data from the BDS B2a Navigation Message. 24-bit CRC has been removed, and the remaining bits of each word are right-shifted. Every word contains 32 bits, and nine words make up one message type. The first word of a message type contains bits 233–264. The second word of a message type contains bits 201–232, and so on. …. The ninth word of a message type contains bits 1–8. For details on data format, see *Figure 15: EPH Data Format – BDS MEO (B2a)*.

Result:

Returns **PAIR001** message and the query result.

Query result message format:

```
$PAIR531,<Status>*<Checksum><CR><LF>
```

Parameter included in the result:

Field	Format	Unit	Description
<Status>	Numeric	-	0 = Failed 1 = Successful

Example:

```
//GPS
$PAIR531,0,0,05,369000,138AF9,101EA3,3338BA,78CBE8,1A60AE,00FFF3,EECD7B,1AF784,2C3D7E,190507,F8C903,141524,1631A1,0D7E3F,60AE7C,002C7A,467A1D,FFF527,62DB0B,16562F,489AEF,FFAD58,1A0AD4*68
$PAIR001,531,1*3D
$PAIR001,531,0*3C
$PAIR531,1*20

//GLONASS
$PAIR531,1,0,03,079AACFF,8305482D,0000009F,17A0921B,8580D948,00000013,80168FB6,A29AB666,00000075,8539BE60,00026277,0000000D,9DD00000,00238000,00000062,0000000D*60
$PAIR001,531,1*3D
$PAIR001,531,0*3C
$PAIR531,1*20

//Galileo
$PAIR531,2,0,10,49E8DAC4,D9680009,DDFEA813,00007E0F,A98274AD,28BD1AFA,D8AE6A95,00000621,FFC67C1A,B401BA14,87130802,00004A6B,280017FF,FD27AFEC,652CDFFF,00003301,1640100C,04FB45AA,0000000B*4C
$PAIR001,531,1*3D
$PAIR001,531,0*3C
$PAIR531,1*20

//BDS GEO
$PAIR531,3,0,01,00075AE1,00F2B9E0,15,2128DC34,3EEE0EFF,55,00000000,01CEA824,15,00F90000,4512440D,55,05B43821,31F1DC00,55,E445CB2B,B0124000,55,F5002D9C,20009540,55,71FC6B48,79FFFFEF,D5,26DC0ACD,0C4875DA,55,F921EAAA,AAAAAAAA,AA*6F
$PAIR001,531,1*3D
$PAIR001,531,0*3C
$PAIR531,1*20

//BDS MEO
$PAIR531,3,0,06,E24057C7,A8001C72,F76059FC,8685FA88,107F813F,800745DF,C803A281,E24097C7,484937E3,D485B7D5,F301EF12,430DE300,9D9F9027,2B9EA499,E240D7C7,61EEC134,2E8A87FC,37FFD194,004AF287,85769D39,0EB28C70*3C
$PAIR001,531,1*3D
$PAIR001,531,0*3C
$PAIR531,1*20

//QZSS
$PAIR531,4,0,02,36A103,800000,000000,000000,000003,D15FCD,00FFFE,FFED74,D13542,1D1532,5C75A1,1FD026,B477FA,2699CA,E91A23,5FCD7C,FCC900,029569,00321D,4E6276,CB7BBF,6D0E44
```

```
,FFDCD5,D1F2FC*6D
$PAIR001,531,1*3D
$PAIR001,531,0*3C
$PAIR531,1*20
```

NOTE

1. GLONASS satellite numbers in NMEA 0183 start from 65. When sending **PAIR531**, you need to align the satellite numbers in NMEA 0183 with the satellite numbers in **PAIR531**, that is, subtract 64 from the GLONASS satellite number in NMEA 0183.
2. QZSS satellite numbers in NMEA 0183 start from 193. When sending **PAIR531**, you need to align the satellite numbers in NMEA 0183 with the satellite numbers in **PAIR531**, that is, subtract 192 from the QZSS satellite number in NMEA 0183.

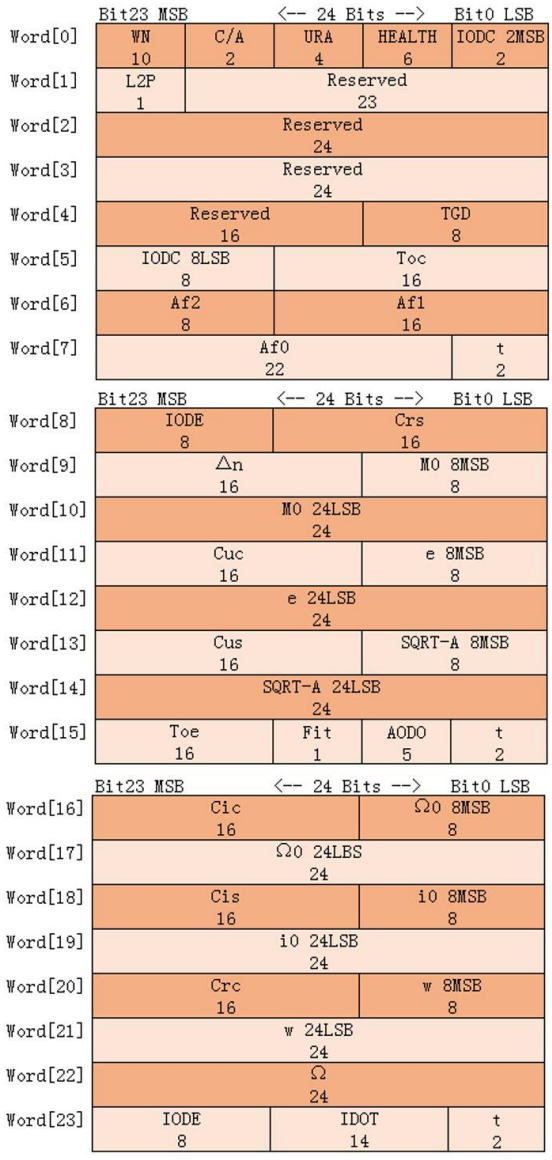


Figure 8: EPH Data Format – GPS (L1) and QZSS (L1)

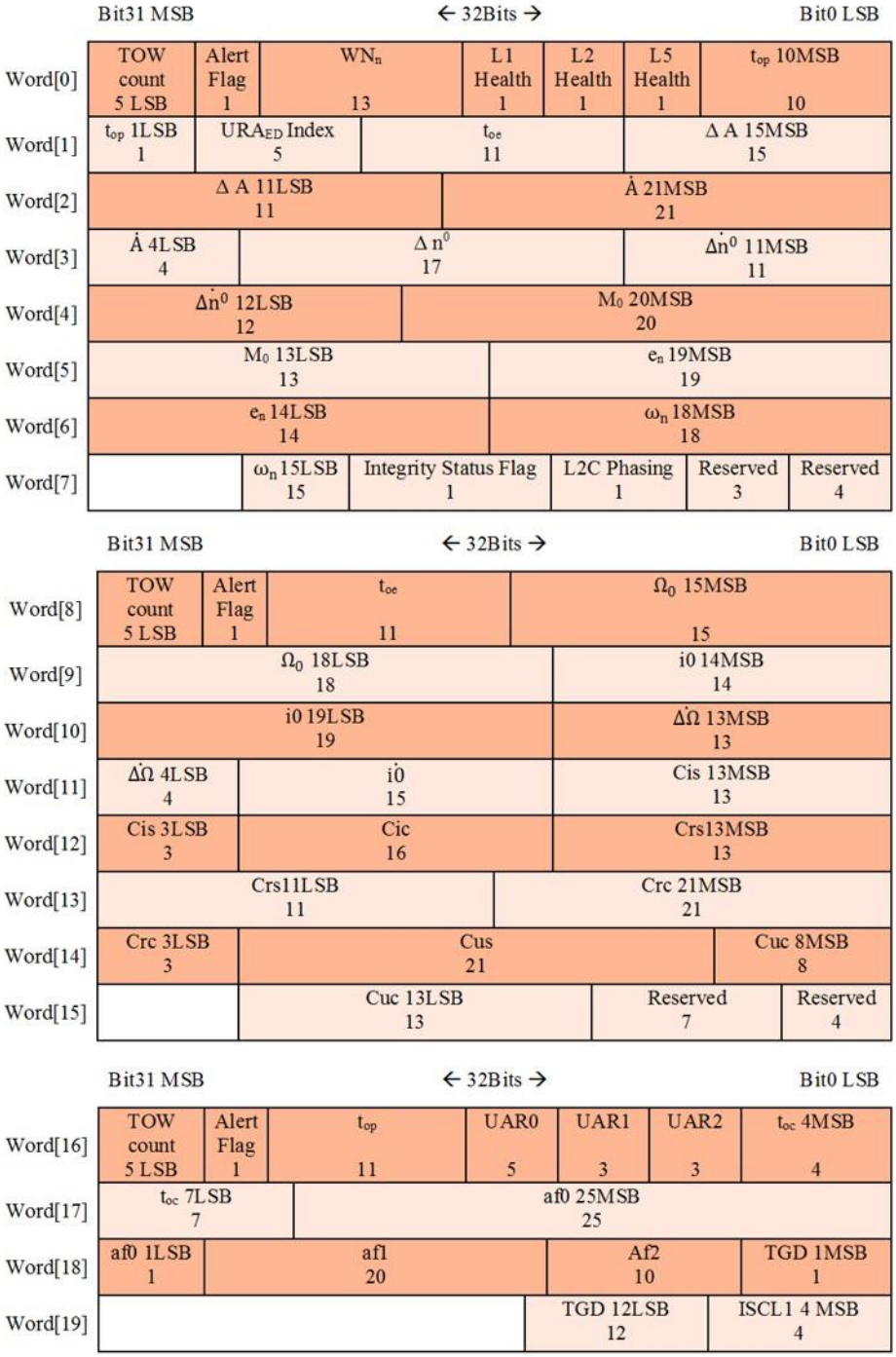


Figure 9: EPH Data Format – GPS (L5) and QZSS (L5)

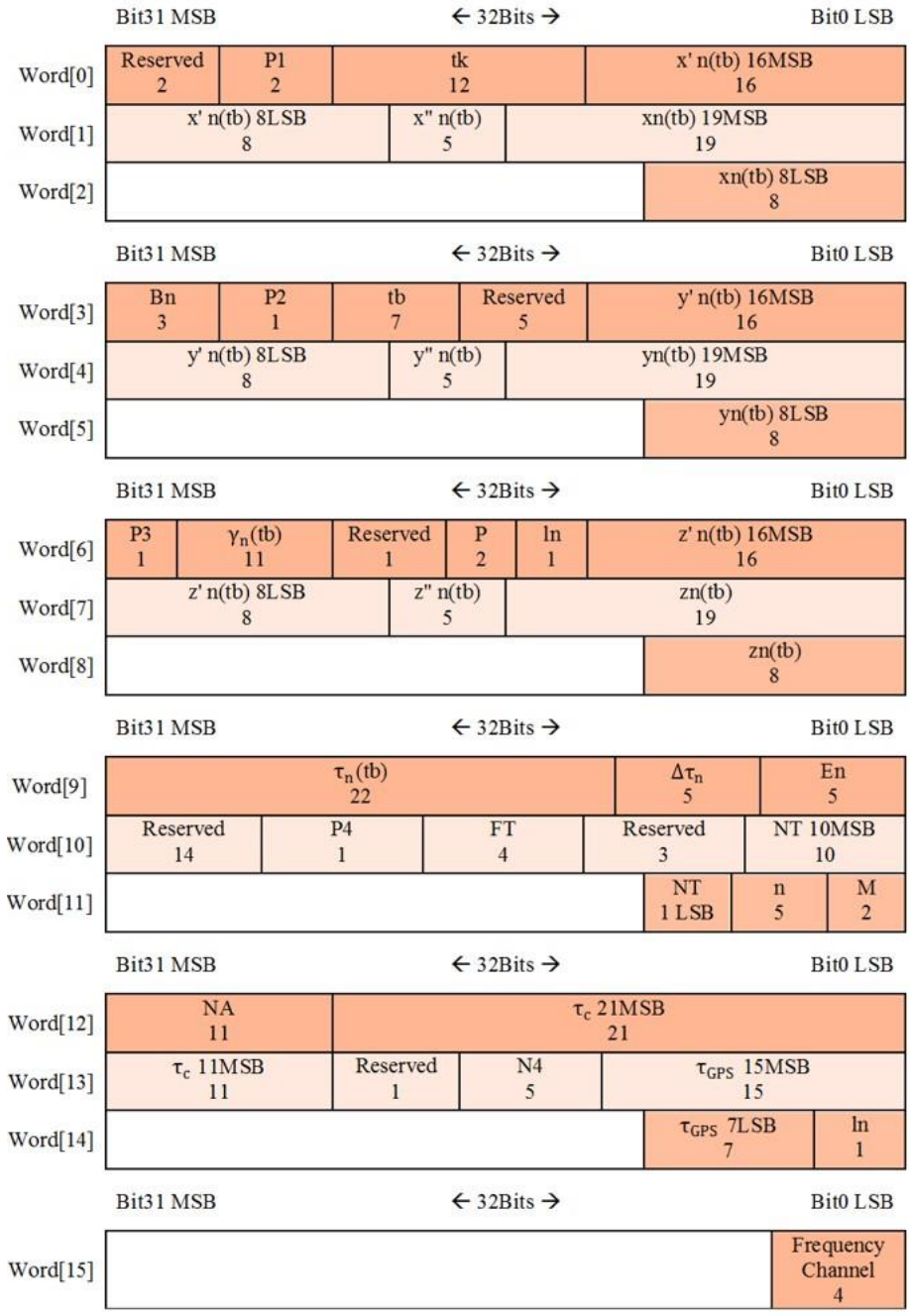


Figure 10: EPH Data Format – GLONASS (L1)

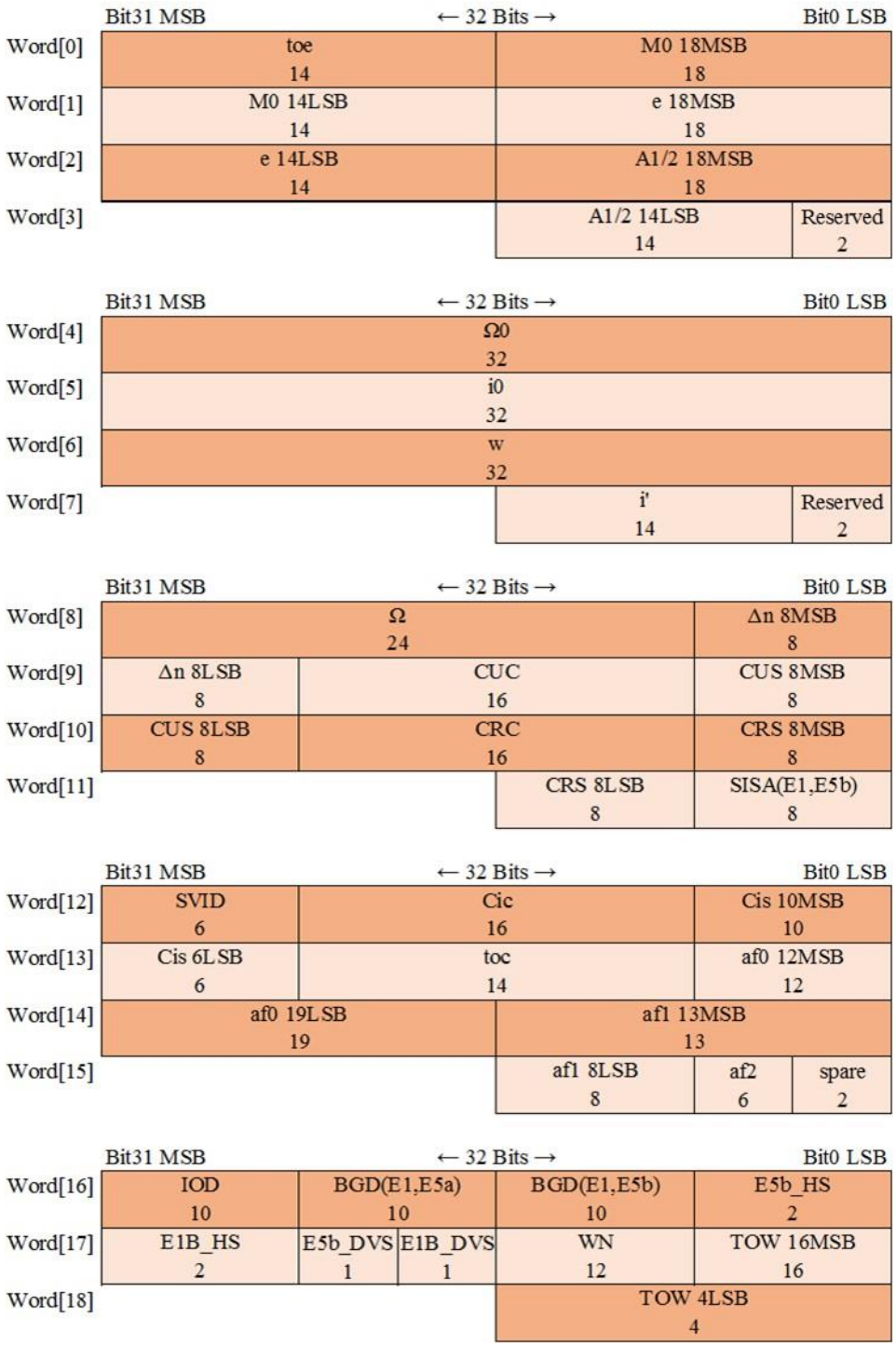


Figure 11: EPH Data Format – Galileo (E1)

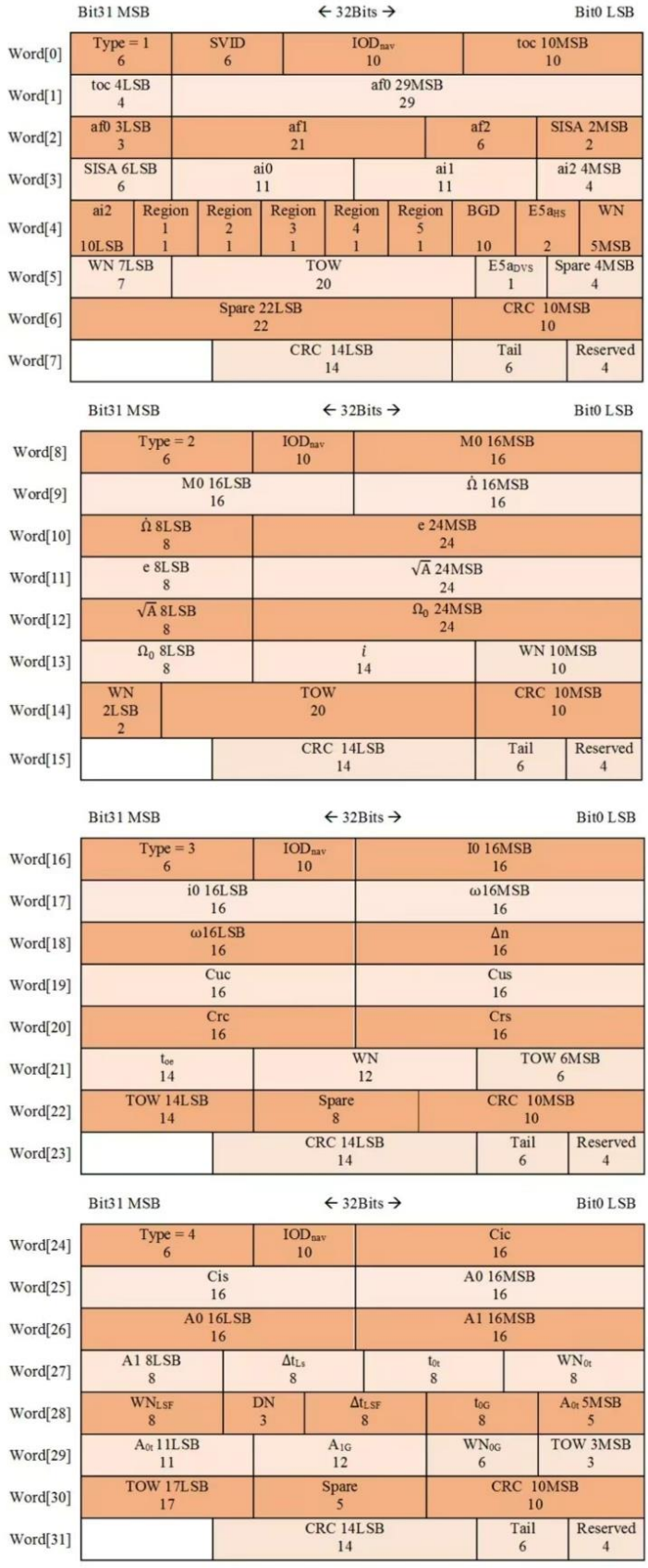


Figure 12: EPH Data Format – Galileo (Ea5)

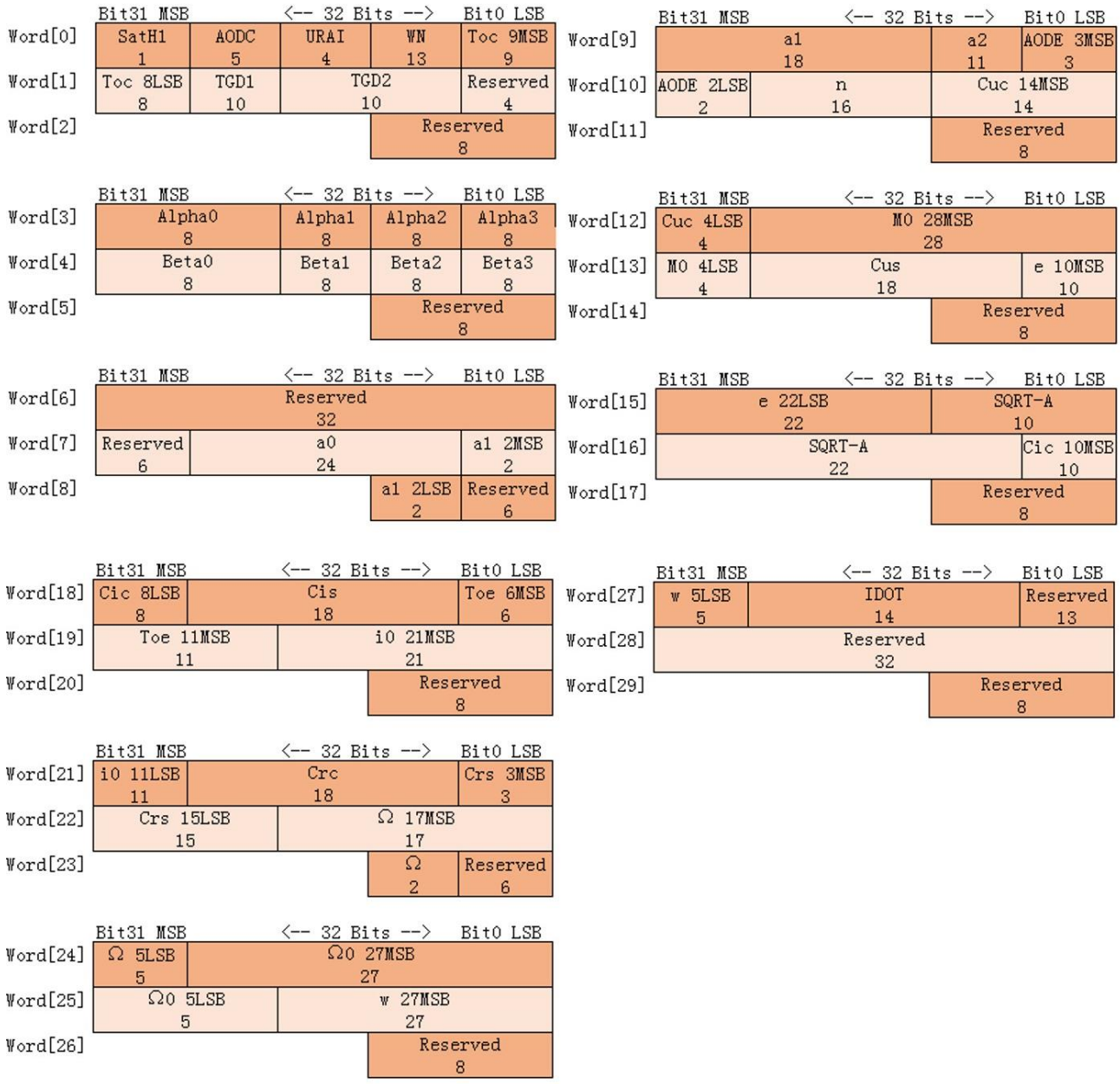


Figure 13: EPH Data Format – BDS GEO (B1I)

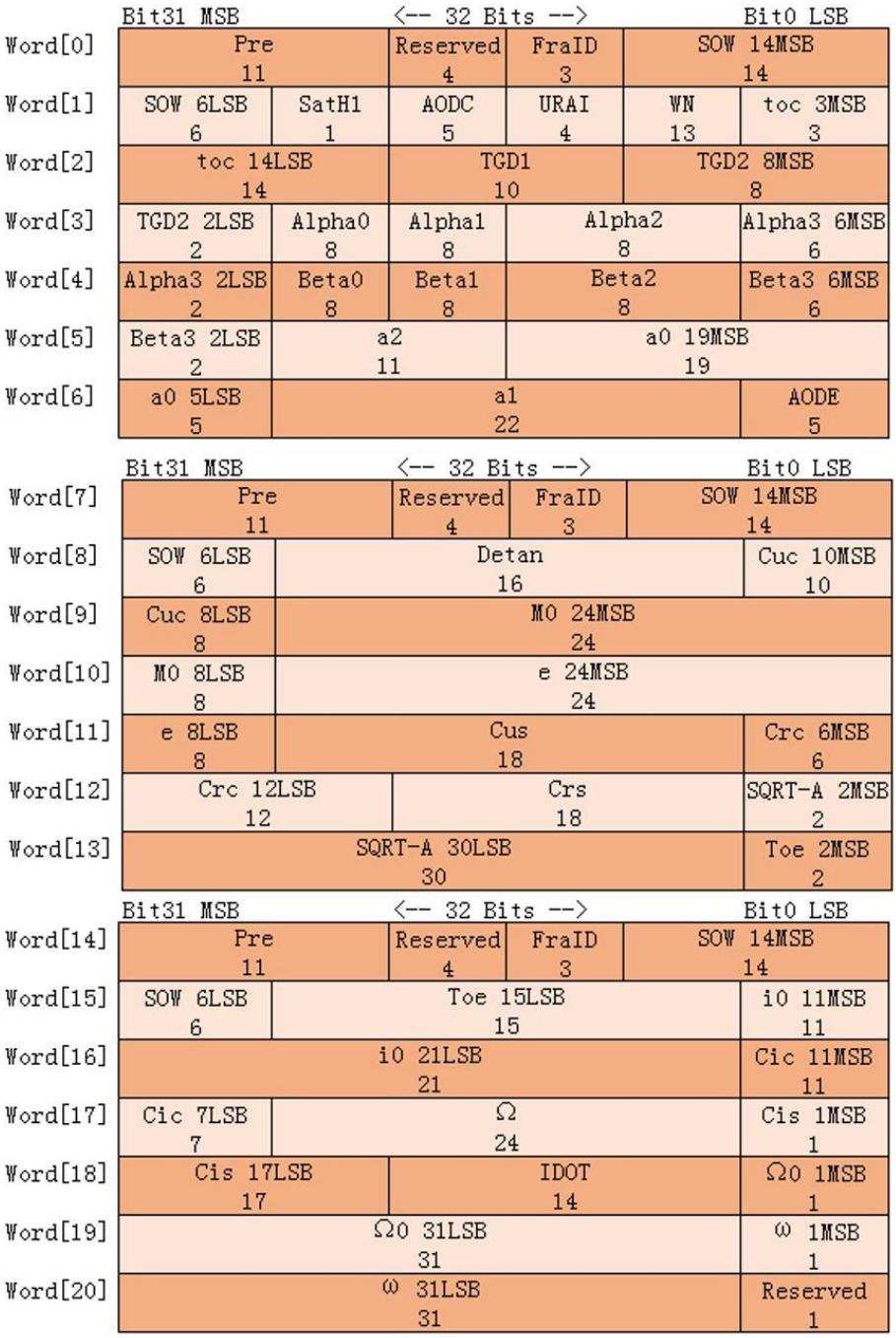


Figure 14: EPH Data Format – BDS MEO (B1I)

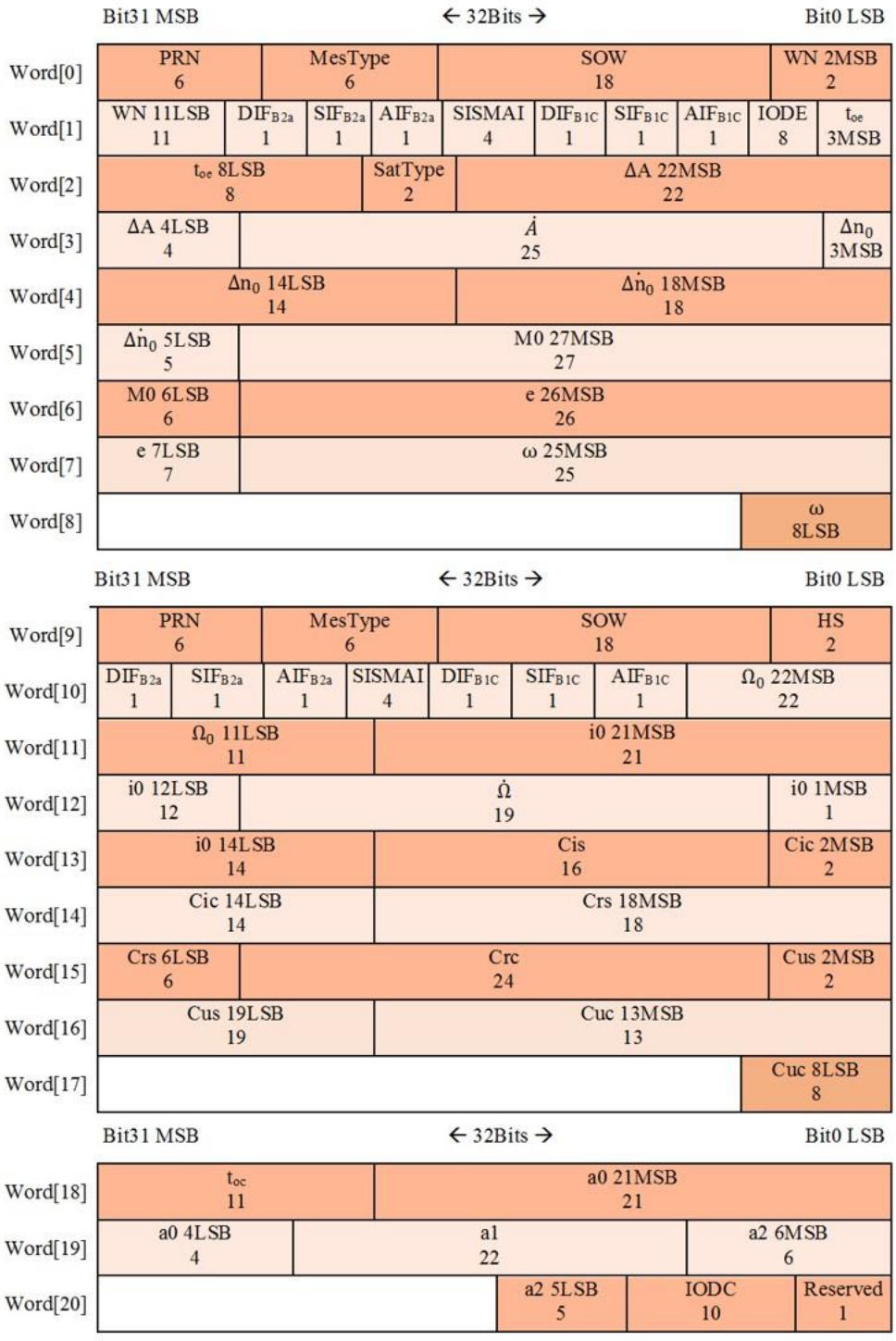


Figure 15: EPH Data Format – BDS MEO (B2a)

3.1.3. PAIR532: PAIR_EPH_GET_DATA

Gets the EPH data for a specified satellite.

Type:

Command

Synopsis:

```
$PAIR532,<Constellation>,<Signal_ID>,<SV_ID>*<Checksum><CR><LF>
```

Parameter:

Field	Format	Unit	Description
<Constellation>	Numeric	-	GNSS System ID. 0 = GPS 1 = GLONASS 2 = Galileo 3 = BDS 4 = QZSS
<Signal_ID>	Numeric	-	Signal ID. 0 = L1 (including Galileo E1 and BDS B1I) 1 = L5 (including GPS/QZSS L5, Galileo E5a and BDS B2a)
<SV_ID>	Numeric	-	SV ID. Ranges for different constellations: GPS: 1–32 GLONASS: 1–24 Galileo: 1–36 BDS: 1–63 (GEO: 1–5, 59–63; MEO: 6–58) QZSS: 1–7

Result:

Returns **PAIR001** message and the query result.

Query result message format:

```
$PAIR532,<Constellation>,<Signal_ID>,<SV_ID>,<EPH_data>*<Checksum><CR><LF>
```

Parameters included in the result:

Field	Format	Unit	Description
<Constellation>	Numeric	-	GNSS System ID. 0 = GPS 1 = GLONASS

Field	Format	Unit	Description
			2 = Galileo 3 = BDS 4 = QZSS
<Signal_ID>	Numeric	-	Signal ID. 0 = L1 1 = L5
<SV_ID>	Numeric	-	SV ID. Ranges for different constellations: GPS: 1–32 GLONASS: 1–24 Galileo: 1–36 BDS: 1–63 (GEO: 1–5, 59–63; MEO: 6–58) QZSS: 1–7
<EPH_Data>	Hexadecimal	-	EPH data. EPH data format for different constellations: **GPS (L1) and QZSS (L1):** Word[0]–Word[23]. 24 words of EPH data from words 3–10 of subframes 1–3 in the GPS or QZSS Navigation Message. The 30-bit raw data of each word have been logically right-shifted by 6 bits to remove the 6 parity bits, leaving 24 bits of valid data per word. For details on data format, see *Figure 8: EPH Data Format – GPS (L1) and QZSS (L1)*. **GPS (L5) and QZSS (L5):** Word[0]–Word[19]. 20 words of the ephemeris data consisting of bits 33–276 from message types 10, 11 and 30 of the Navigation Message. Every word contains 32 bits, and eight words make up one message type. The last four words contain the clock data. For details on data format, see *Figure 9: EPH Data Format – GPS (L5) and QZSS (L5)*. **GLONASS (L1):** Word[0]–Word[15]. Word[0]–Word[14] of EPH data are from strings 1–5 of a frame in a GLONASS Navigation Message. Word[15] indicates the frequency channel (range: 1–14). For details on data format, see *Figure 10: EPH Data Format – GLONASS (L1)*. **Galileo (E1):** Word[0]–Word[18]. 19 words of EPH data in the Galileo Navigation Message include 16 words from the subframes 1–4, and 3 words from

Field	Format	Unit	Description
			subframe 5, which include with BGD, signal health status, data validity, and GST. Note that only Word[0]–Word[15] need to be input when sending ephemeris, while Word[0]–Word[18] will be provided when receiving ephemeris. The Page Type (6-bit) and IOD_{nav} (10-bit) fields have been removed and the remaining bits of each word are right-shifted. For details on data format, see *Figure 11: EPH Data Format – Galileo (E1)*. **Galileo (E5a):** Word[0]–Word[31]. 32 words of ephemeris data from word types 1–4 in subframes of the Galileo E5a Navigation Message. Every word contains 32 bits, and eight words make up one word type. The first word of a word type contains bits 217–248. The second word of a word type contains bits 185–216, and so on. …. The eighth word of a word type contains bits 1–24. For details on data format, see *Figure 12: EPH Data Format – Galileo (Ea5)*. **BDS GEO (B1I):** Word[0]–Word[29]. 30 words of EPH data from pages 1–10 of subframe 1 in the BDS GEO D2 Navigation Message. The 30-bit raw data of each word have been logically right-shifted by 8 bits to remove the 8 parity bits. For details on data format, see *Figure 13: EPH Data Format – BDS GEO (B1I)*. **BDS MEO (B1I):** Word[0]–Word[20]. 21 words of EPH data from subframes 1–3 in the BDS MEO D1 Navigation Message. The 30-bit raw data of each word have been logically right-shifted by 8 bits to remove the 8 parity bits. For details on data format, see *Figure 14: EPH Data Format – BDS MEO (B1I)*. **BDS MEO (B2a):** Word[0]–Word[20]. 21 words of ephemeris subframe data from the BDS B2a Navigation Message.

Field	Format	Unit	Description
			24-bit CRC has been removed, and the remaining bits of each word are right-shifted. Every word contains 32 bits, and nine words make up one message type. The first word of a message type contains bits 233–264. The second word of a message type contains bits 201–232, and so on. …. The ninth word of a message type contains bits 1–8. For details on data format, see *Figure 15: EPH Data Format – BDS MEO (B2a)*.

Example:

```
//GPS
$PAIR532,0,0,8*2A
$PAIR001,532,1*3E
$PAIR001,532,0*3F
$PAIR532,0,0,08,3E5000,72962C,B66E4F,B1894A,86D40B,2B7080,00FFF0,EAA8AB,2B0771,33CE27,3ABE45,06A904,B7489D,0748A1,0CA1E6,70807F,FFE387,4CC730,004B26,EDAA61,267D0C,52705A,FFA33F,2BFD9F*61

//GLONASS
$PAIR532,1,0,10*12
$PAIR001,532,1*3E
$PAIR001,532,0*3F
$PAIR532,1,0,10,0521129B,92166C00,000000E4,15208EF0,311172BE,0000001F,C046B222,180D543C,000000E4,08656C00,0002A01C,000000A9,07300000,01A40000,00000020,00000001*16

//Galileo
$PAIR532,2,0,10*11
$PAIR001,532,1*3E
$PAIR001,532,0*3F
$PAIR532,2,0,10,4B29379F,7BD80009,90C2A813,000089D3,A9823149,28BD1CC8,D9B6AF27,00000619,FFC67C1A,1901A414,D2124102,0000696B,280013FF,FD2CAFEC,650E3FFF,00003301,1840100C,04FB46E9,00000007*4C

//BDS GEO
$PAIR532,3,0,3*22
$PAIR001,532,1*3E
$PAIR001,532,0*3F
$PAIR532,3,0,03,00075AE1,0005FAF0,15,2128DC34,3EEE0EFF,55,00000000,03CF3F80,15,1C02400
```

```
0,446B81F4,55,2E86E526,B2045801,55,E57F9B2B,AD088801,55,3000501C,20009517,55,B9DC4E08
,36CDFFF1,55,2ED49A12,2F25A36E,55,111D6AAA,AAAAAAAA,AA*18

//BDS MEO
$PAIR532,3,0,16*16
$PAIR001,532,1*3E
$PAIR001,532,0*3F
$PAIR532,3,0,16,E24051BA,E8001D7A,3283E50C,4B7CBF81,8D807DC4,0006B895,A7F3EBC1,E240
91BA,8822EFE8,28E3BB4E,46037EDE,EE2BD0B9,CC6FB32F,2B9E8FAD,E240D1BA,A0650139,BB
2CBFFF,0FFFD986,0034FEEC,3E4FAE87,46D72F64*42

//QZSS
$PAIR532,4,0,2*24
$PAIR001,532,1*3E
$PAIR001,532,0*3F
$PAIR532,4,0,02,3EE107,800000,000000,000000,000003,654812,00FFFD,FFCABC,651777,07C1B6,4
401CD,0DFE26,6AF306,F6C2CA,EB5349,48127E,03295B,CAB930,F78F1C,FDCB97,269EBF,D7368F
,FFFA98,651E36*65
```

3.1.4. PAIR533: PAIR_EPH_CLEAR

Clears the EPH data in the critical memory area.

Type:

Command

Synopsis:

```
$PAIR533*<Checksum><CR><LF>
```

Parameter:

None

Result:

Returns **PAIR001** message.

Example:

```
$PAIR533*3F
$PAIR001,533,0*3E
```

3.1.5. PAIR534: PAIR_EPH_NOTIFY_ENABLE

Enables/disables the notification that updated EPH data is available.

Type:

Command

Synopsis:

```
$PAIR534,<Enabled>*<Checksum><CR><LF>
```

Parameter:

Field	Format	Unit	Description
<Enabled>	Numeric	-	Enable/Disable the notification that updated EPH data is available. 0 = Disable 1 = Enable

Result:

Returns **PAIR001** message.

Example:

```
$PAIR534,1*25
$PAIR001,534,0*39
```

3.1.6. PAIR535: PAIR_EPH_NOTIFY

Outputs the notification that updated EPH data is available.

Type:

Output

Synopsis:

```
$PAIR535,<Constellation>,<Signal_ID>,<SV_Information>*<Checksum><CR><LF>
```

Parameter:

Field	Format	Unit	Description
<Constellation>	Numeric	-	GNSS System ID. 0 = GPS 1 = GLONASS

Field	Format	Unit	Description
			2 = Galileo 3 = BDS 4 = QZSS
<Signal_ID>	Numeric	-	Signal ID. 0 = L1 (including Galileo E1 and BDS B1I) 1 = L5 (including GPS/QZSS L5, Galileo E5a and BDS B2a)
<SV_Information>	Hexadecimal	-	SV ID bitmap to indicate the satellite(s) whose updated EPH data are available. Length: 64 bits.

Example:

$PAIR535,0,0,0000000010080000*1C
//The binary format of 0000000010080000 is: 0001 0000 0000 1000 0000 0000 0000 0000. The corresponding satellite numbers are 20 and 29.

3.2. ALM Related Messages

3.2.1. PAIR550: PAIR_ALM_GET_STATUS

Queries satellites whose valid ALM data will be available for a specified future time period.

Type:

Command

Synopsis:

$PAIR550,<Constellation>,<Time_Interval>*<Checksum><CR><LF>

Parameter:

Field	Format	Unit	Description
<Constellation>	Numeric	-	GNSS System ID. 0 = GPS 1 = GLONASS 2 = Galileo 3 = BDS 4 = QZSS
<Time_Interval>	Numeric	Day	Specified future time period. Range: 1–91.

Result:

Returns **PAIR001** message and the query result.

Query result message format:

```
$PAIR550,<Constellation>,<L1_SV>[,<Midi_SV>]*<Checksum><CR><LF>
```

Parameters included in the result:

Field	Format	Unit	Description
<Constellation>	Numeric	-	GNSS System ID. 0 = GPS 1 = GLONASS 2 = Galileo 3 = BDS 4 = QZSS
<L1_SV>	Hexadecimal	-	Valid ALM SV ID (L1 band). Field Lengths for different constellations (in ASCII format) are as follows. GPS: 8 bytes GLONASS: 6 bytes Galileo: 16 bytes BDS: 16 bytes QZSS: 8 bytes Note that each bit represents a satellite, and SV ID counting starts from the least significant bit. Use GPS as an example. If Bit 0 is 1, the ALM data of GPS satellite 1 is available.
<Midi_SV>	Hexadecimal	-	Valid ALM SV ID (L5 band). When **<Constellation>** = 1, this field is omitted. Field Lengths for different constellations (in ASCII format) are as follows. GPS: 8 bytes GLONASS: 6 bytes Galileo: 16 bytes BDS: 16 bytes QZSS: 8 bytes Note that each bit represents a satellite, and SV ID counting starts from the least significant bit. Use GPS as an example. If Bit 0 is 1, the ALM data of GPS satellite 1 is available.

Example:

```
$PAIR550,4,90*07
```

$PAIR001,550,1*3A
$PAIR001,550,0*3B
$PAIR550,4,0000004E,0000004E*22
//The binary format of 0000004E is 0000 0000 0000 0000 0000 0000 0100 1110. The corresponding satellite numbers are 2, 3, 4, 7.

3.2.2. PAIR551: PAIR_ALM_SET_DATA

Sends ALM data to GNSS module.

Type:

Input

Synopsis:

$PAIR551,<Constellation>,<Signal_ID>,<SV_ID>,<WeekNo>,<ALM_Data>*<Checksum><CR><LF>

Parameter:

Field	Format	Unit	Description
<Constellation>	Numeric	-	GNSS System ID. 0 = GPS 1 = GLONASS 2 = Galileo 3 = BDS 4 = QZSS
<Signal_ID>	Numeric	-	Signal ID. 0 = L1 (including Galileo E1 and BDS B1I) 1 = L5 (including GPS/QZSS L5, Galileo E5a and BDS B2a)
<SV_ID>	Numeric	-	SV ID. Ranges for different constellations: GPS: 1–32 GLONASS: 1–24 Galileo: 1–36 BDS: 1–63 (GEO: 1–5, 59–63; MEO: 6–58) QZSS: 1–7
<WeekNo>	Hexadecimal	-	Almanac reference week number.
<ALM_Data>	Hexadecimal	-	ALM data. ALM data format for different constellations: **GPS (L1) and QZSS (L1):** Word[0]–Word[7]. 8 words of ALM data from pages 1–24 of subframe 5, as well as pages 2–5 and 7–9 of subframe 4 in the GPS or QZSS Navigation Message. The 30-bit raw data of each word have been logically right-

Field	Format	Unit	Description
			shifted by 6 bits to remove the 6 parity bits, leaving 24 bits of valid data per word. For details on data format, see *Figure 16: ALM Data Format – GPS (L1) and QZSS (L1)*. **GPS (L5) and QZSS (L5):** Word[0]–Word[5]. 6 words of the almanac data from message type 37 in the GPS or QZSS Navigation Message. Every word contains 24 bits. For details on data format, see *Figure 17: ALM Data Format – GPS (L5) and QZSS (L5)*. **GLONASS (L1):** Word[0]–Word[5]. 6 words of ALM data from strings 6–15 of the GLONASS Navigation Message. For details on data format, see *Figure 18: ALM Data Format – GLONASS (L1)*. **Galileo (E1):** Word[0]–Word[3]. 4 words of ALM data from subframes 7–10 in the Galileo Navigation Message. For details on data format, see *Figure 19: ALM Data Format – Galileo (E1)*. **Galileo (E5a):** Word[0]–Word[3]. 4 words of the almanac data from word types 5 and 6 in the Galileo Navigation Message. Every word contains 32 bits, and 4 words make up almanac data of one satellite. For details on data format, see *Figure 20: ALM Data Format – Galileo (E5a)*. **BDS (B1I):** Word[0]–Word[6]. 7 words of ALM data from pages 37–60 and 95–100 in subframe 5 (for GEO), pages 1–24 in subframe 4, and pages 1–6 in subframe 5 (for MEO). 11 parity bits have been removed. For details on data format, see *Figure 21: ALM Data Format – BDS (B1I)*. **BDS MEO (B2a):** Word[0]–Word[8]. 9 words of the almanac data from message type 40 of the BDS Navigation Message. Every word contains 32 bits. For Word[8], only 8 bits are valid. For details on data format, see *Figure 22: ALM Data Format – BDS (B2a)*.

Result:

Returns **PAIR001** message and the query result.

Query result message format:

```
$PAIR551,<Status>*<Checksum><CR><LF>
```

Parameter included in the result:

Field	Format	Unit	Description
<Status>	Numeric	-	0 = Failed 1 = Successful

Example:

```
//GPS
$PAIR551,0,0,02,08F9,4285A8,90104B,FD3F00,A10EC7,AFD145,CCB416,3569A4,BE0058*30
$PAIR001,551,1*3B
$PAIR001,551,0*3A
$PAIR551,1*26

//GLONASS
$PAIR551,1,0,02,0012,16C908E0,0169A56C,000000A2,97EDF638,42502DE6,000000DD*35
$PAIR001,551,1*3B
$PAIR001,551,0*3A
$PAIR551,1*26

//Galileo
$PAIR551,2,0,02,04FB,0080221A,5FF94A91,BFC06E34,00D00020*47
$PAIR001,551,1*3B
$PAIR001,551,0*3A
$PAIR551,1*26

//BDS
$PAIR551,3,0,01,03AD,E2411C20,48072BAF,E40039DE,32B1F01C,48911188,00E60244,D55D5F97*69
$PAIR001,551,1*3B
$PAIR001,551,0*3A
$PAIR551,1*26

//QZSS
$PAIR551,4,0,02,08FA,C27AD8,15D037,FF3B01,CAEA53,60CCA9,BFC206,43CC9F,FF001C*4A
$PAIR001,551,1*3B
$PAIR001,551,0*3A
```

$PAIR551,1*26

NOTE

1. GLONASS satellite numbers in NMEA 0183 start from 65. When sending **PAIR551**, you need to align the satellite numbers in NMEA 0183 with the satellite numbers of **PAIR551**, that is, subtract 64 from the GLONASS satellite number in NMEA 0183.
2. QZSS satellite numbers in NMEA 0183 start from 193. When sending **PAIR551**, you need to align the satellite numbers in NMEA 0183 with the satellite numbers of **PAIR551**, that is, subtract 192 from the QZSS satellite number in NMEA 0183.

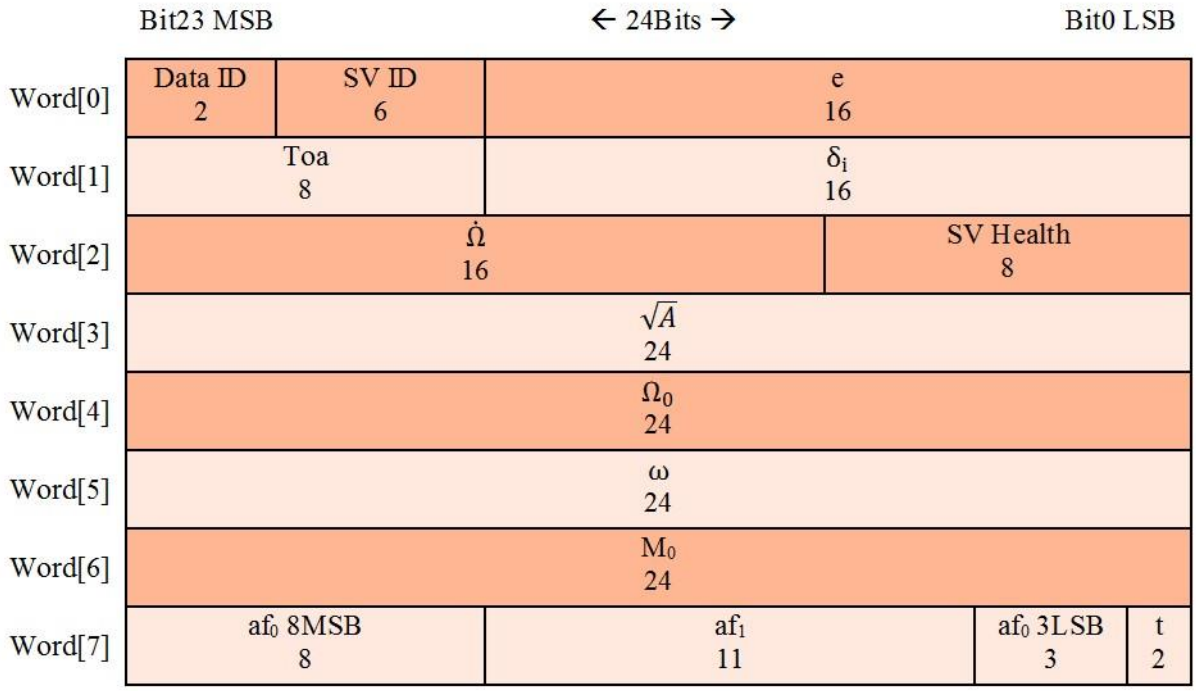


Figure 16: ALM Data Format – GPS (L1) and QZSS (L1)

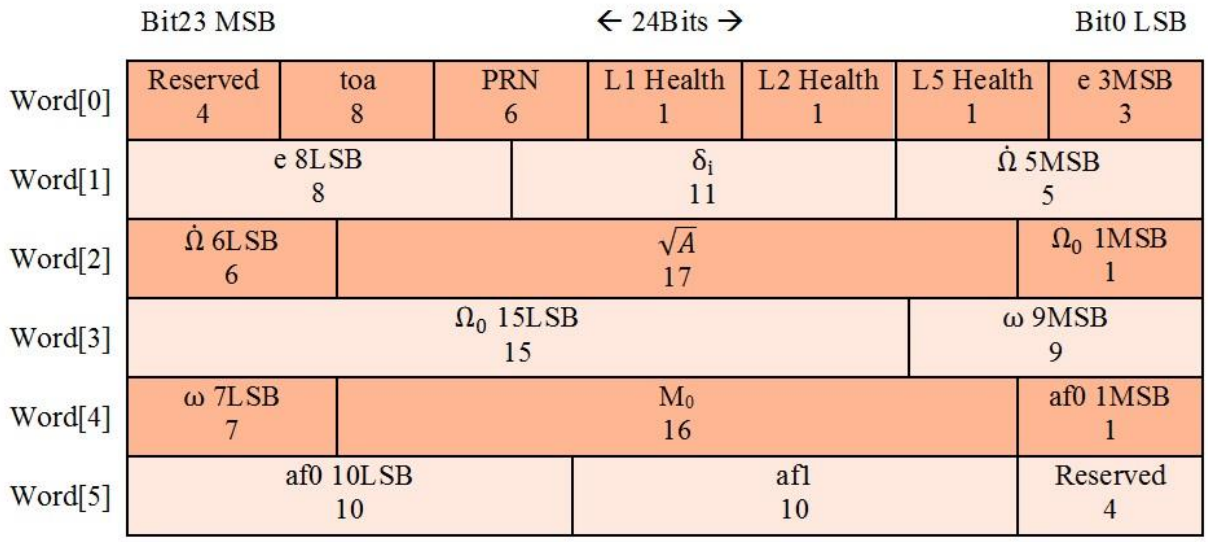


Figure 17: ALM Data Format – GPS (L5) and QZSS (L5)

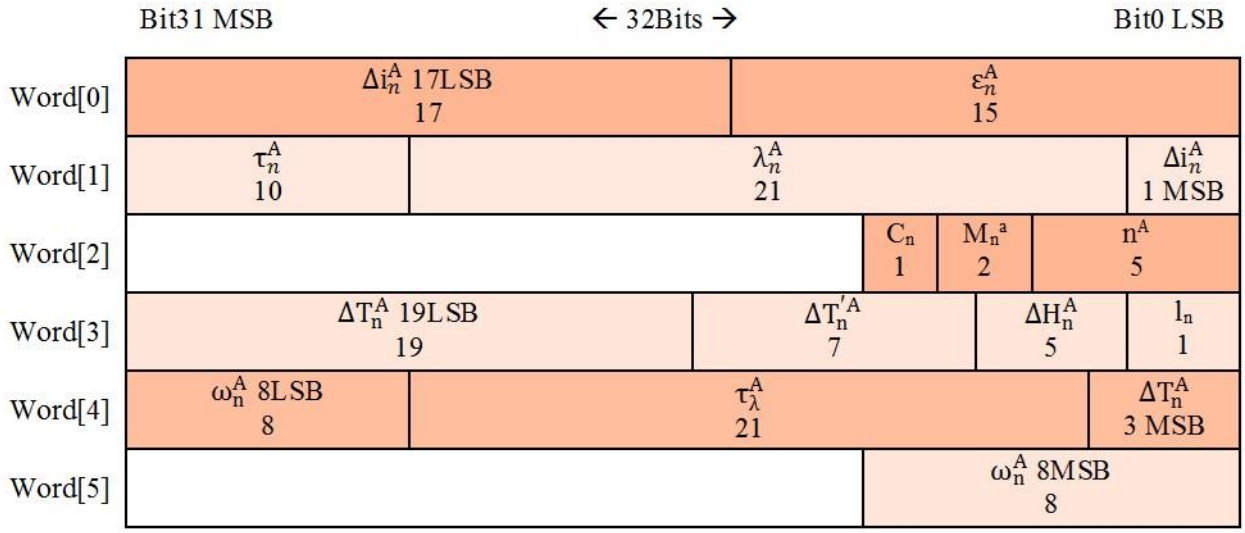


Figure 18: ALM Data Format – GLONASS (L1)

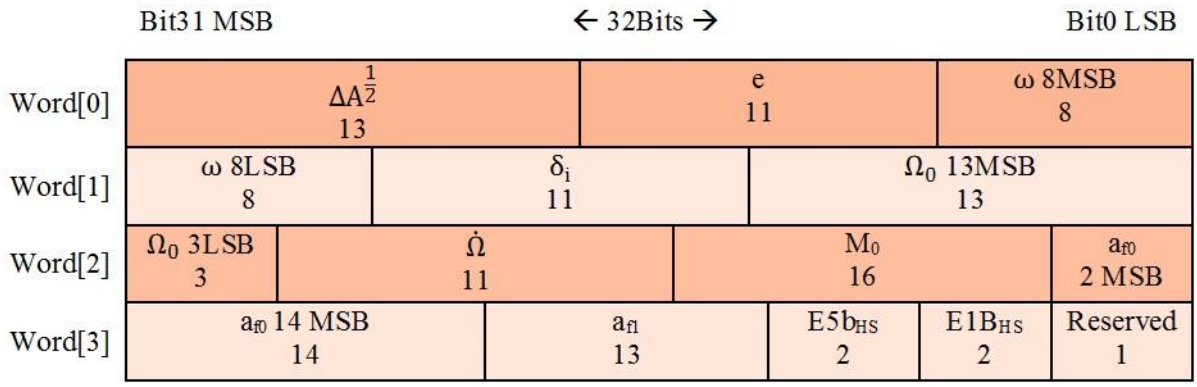


Figure 19: ALM Data Format – Galileo (E1)

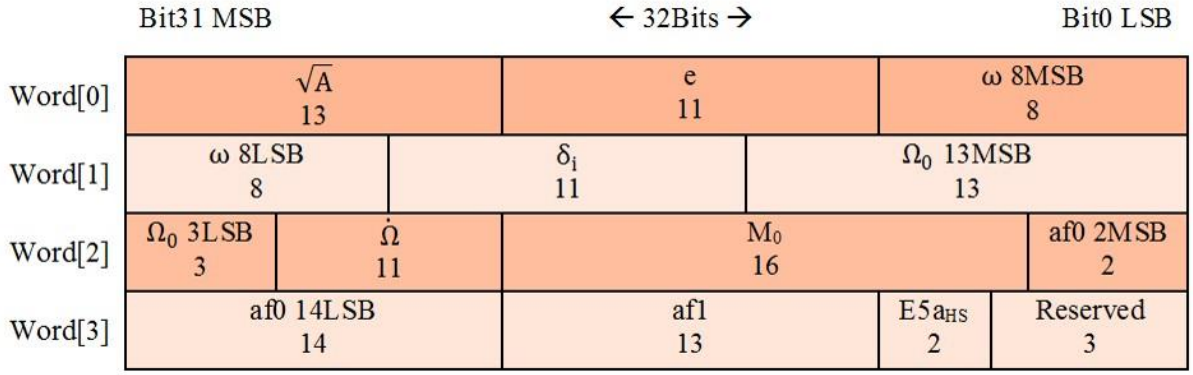


Figure 20: ALM Data Format – Galileo (E5a)

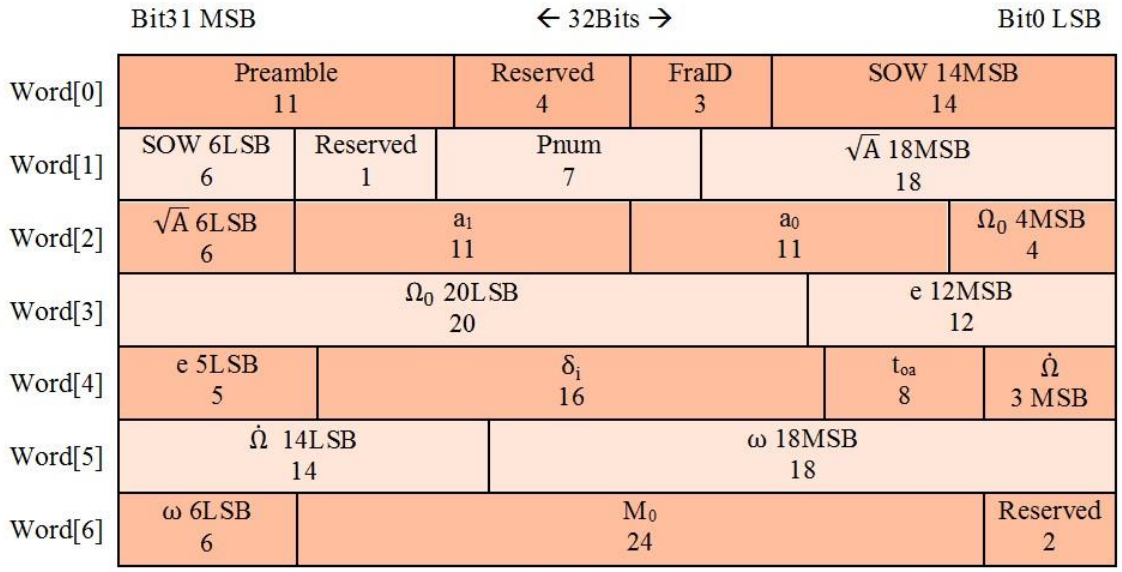


Figure 21: ALM Data Format – BDS (B1I)

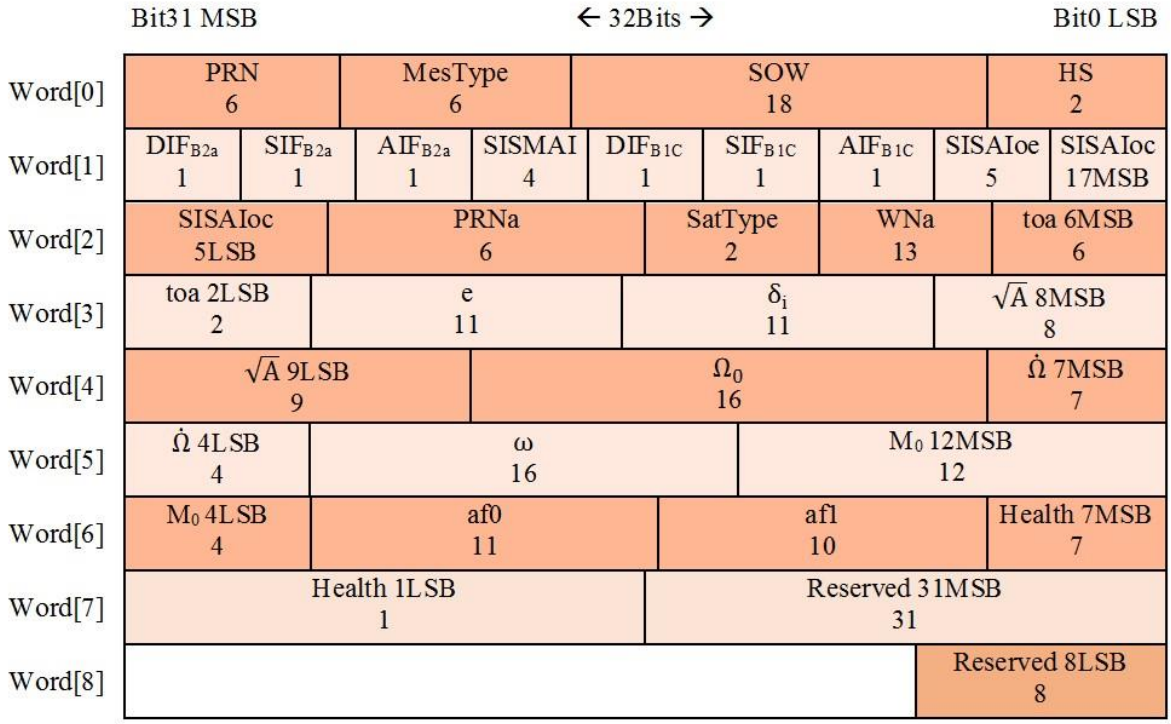


Figure 22: ALM Data Format – BDS (B2a)

3.2.3. PAIR552: PAIR_ALM_GET_DATA

Gets the ALM data of the specified satellite.

Type:

Command

Synopsis:

```
$PAIR552,<Constellation>,<Signal_ID>,<SV_ID>*<Checksum><CR><LF>
```

Parameter:

Field	Format	Unit	Description
<Constellation>	Numeric	-	GNSS System ID. 0 = GPS 1 = GLONASS 2 = Galileo 3 = BDS 4 = QZSS
<Signal_ID>	Numeric	-	Signal ID. 0 = L1 (including Galileo E1 and BDS B1I)

Field	Format	Unit	Description
			1 = L5 (including GPS/QZSS L5, Galileo E5a and BDS B2a)
<SV_ID>	Numeric	-	SV ID. Ranges for different constellations: GPS: 1–32 GLONASS: 1–24 Galileo: 1–36 BDS: 1–63 (GEO: 1–5, 59–63; MEO: 6–58) QZSS: 1–7

Result:

Returns **PAIR001** message and the query result.

Query result message format:

```
$PAIR552,<Constellation>,<Signal_ID>,<SV_ID>,<WN>,<ALM_Data>*<Checksum><CR><LF>
```

Parameters included in the result:

Field	Format	Unit	Description
<Constellation>	Numeric	-	GNSS System ID. 0 = GPS 1 = GLONASS 2 = Galileo 3 = BDS 4 = QZSS
<Signal_ID>	Numeric	-	Signal ID. 0 = L1 (including Galileo E1 and BDS B1I) 1 = L5 (including GPS/QZSS L5, Galileo E5a and BDS B2a)
<SV_ID>	Numeric	-	SV ID. Ranges for different constellations: GPS: 1–32 GLONASS: 1–24 Galileo: 1–36 BDS: 1–63 (GEO: 1–5, 59–63; MEO: 6–58) QZSS: 1–7
<WN>	Hexadecimal	Week	Almanac reference week number.
<ALM_Data>	Hexadecimal	-	ALM data. ALM data format for different constellations: **GPS (L1) and QZSS (L1):** Word[0]–Word[7]. 8 words of ALM data from pages 1–24 of subframe 5, as well as pages 2–5 and 7–9 of subframe 4 in the GPS or QZSS Navigation Message. The 30-bit raw data of each word have been logically right-

Field	Format	Unit	Description
			shifted by 6 bits to remove the 6 parity bits, leaving 24 bits of valid data per word. For details on data format, see *Figure 16: ALM Data Format – GPS (L1) and QZSS (L1)*. **GPS (L5) and QZSS (L5):** Word[0]–Word[5]. 6 words of the almanac subframe data from message type 37 in the GPS or QZSS Navigation Message. Every word contains 24 bits. For details on data format, see *Figure 17: ALM Data Format – GPS (L5) and QZSS (L5)*. **GLONASS (L1):** Word[0]–Word[5]. 6 words of ALM data from strings 6–15 of the GLONASS Navigation Message. For details on data format, see *Figure 18: ALM Data Format – GLONASS (L1)*. **Galileo (E1):** Word[0]–Word[3]. 4 words of ALM data from subframes 7–10 in the Galileo Navigation Message. For details on data format, see *Figure 19: ALM Data Format – Galileo (E1)*. **Galileo (E5a):** Word[0]–Word[3]. 4 words of the almanac subframe data from word types 5 and 6 in the Galileo Navigation Message. Every word contains 32 bits, and 4 words make up almanac data of one satellite. For details on data format, see *Figure 20: ALM Data Format – Galileo (E5a)*. **BDS (B1I):** Word[0]–Word[6]. 7 words of ALM data from pages 37–60 and 95–100 in subframe 5 (for GEO), pages 1–24 in subframe 4, and pages 1–6 in subframe 5 (for MEO). 11 parity bits have been removed. For details on data format, see *Figure 21: ALM Data Format – BDS (B1I)*. **BDS MEO (B2a):** Word[0]–Word[8]. 9 words of the almanac data from message type 40 of the BDS Navigation Message. Every word contains 32 bits. For Word[8], only 8 bits are valid. For details on data format, see *Figure 22: ALM Data Format – BDS (B2a)*.

Example:

```
//GPS
$PAIR552,0,0,8*2C
$PAIR001,552,1*38
$PAIR001,552,0*39
$PAIR552,0,0,08,08FA,484B97,0F0876,FD4C00,A10C42,82533D,0C60FC,EC2109,EA0014*42

//GLONASS
$PAIR552,1,0,16*12
$PAIR001,552,1*38
$PAIR001,552,0*39
$PAIR555,1,0,16,0012,13A98271,81948442,000000B0,9802F37E,BA2092F6,000000FB*4A

//Galileo
$PAIR552,2,0,10*17
$PAIR001,552,1*38
$PAIR001,552,0*39
$PAIR552,2,0,10,04FB,006801DB,700EB530,5FC9613B,FB1BFFE0*49

//BDS
$PAIR552,3,0,16*10
$PAIR001,552,1*38
$PAIR001,552,0*39
$PAIR552,3,0,16,03AD,E2415C22,64D32B99,BBFD6D62,90F8F1BB,7065A18F,FD7E8C62,BF57867D
*1D

//QZSS
$PAIR552,4,0,2*22
$PAIR001,552,1*38
$PAIR001,552,0*39
$PAIR552,4,0,02,08FB,C27ADC,7FCFD6,FF1E01,CAEAD7,5BC3FA,BFDD50,52BA0D,FF0019*4D
```

3.2.4. PAIR553: PAIR_ALM_CLEAR

Clears the ALM data in the critical memory area.

Type:

Command

Synopsis:

```
$PAIR553*<Checksum><CR><LF>
```

Parameter:

None

Result:

Returns **PAIR001** message.

Example:

```
$PAIR553*39
$PAIR001,553,0*38
```

4 Appendix A References

Table 9: Related Documents

Document Name
[1] Quectel_LC29H&LC79H_Series_GNSS_Protocol_Specification
[2] Quectel_LC29H_Series_Hardware_Design
[3] Quectel_LC79H_Series_Hardware_Design
[4] Quectel_LC29H&LC79H_Series_AGNSS_Application_Note

Table 10: Terms and Abbreviations

Abbreviation	Description
AGNSS	Assisted GNSS (Global Navigation Satellite System)
ALM	Almanac
ALP	Adaptive Low Power
BDS	BeiDou Navigation Satellite System
BGD	Broadcast Group Delay
CPU	Central Processing Unit
CRC	Cyclic Redundancy Check
DSP	Digital Signal Processor
EPH	Ephemeris
EPO	Extended Prediction Orbit
FLP	Fitness Low Power
GEO	Geostationary Orbit

Abbreviation	Description
GLP	GPS Low Power
GNSS	Global Navigation Satellite System
GPIO	General-Purpose Input/Output
GPS	Global Positioning System
GST	Galileo System Time
ID	Identifier
IOD	Issue of Data
LSB	Least Significant Bit
ME	Measurement Engine
MEO	Medium Earth Orbit
MSB	Most Significant Bit
NavIC	Indian Regional Navigation Satellite System
NMEA	NMEA (National Marine Electronics Association) 0183 Interface Standard
PE	Positioning Engine
QZSS	Quasi-Zenith Satellite System
RAM	Random Access Memory
RF	Radio Frequency
RTC	Real-Time Clock
SRAM	Static Random Access Memory
SV	Satellites in View
TCXO	Temperature Compensated Crystal Oscillator
TTFF	Time to First Fix

5 Appendix B Special Characters

Table 11: Special Characters

Special Character	Definition
<CR>	Carriage return character.
<LF>	Line feed character.
<...>	Parameter name. Angle brackets do not appear in the message.
[...]	Optional field of a message. Square brackets do not appear in the message.
{…}	Repeated field of a message. Curly brackets do not appear in the message.
Underline	Default setting of a parameter.